

Shotton Mill Project

Plant & Asset Numbering Procedure

SML Project Standard No.21 Rev 2.1

Prepared by :

Rob McMahon



SHOTTON MILL LIMITED	SHOTTON MILL PROJECT PLANT & ASSET NUMBERING PROCEDURE	Date: 12.09.2023
		Rev: 2.1
		Document Number: MK-SHO-SMP-SML- PRM-045-PRC-003

PROJECT STANDARD NO: 21 – Rev 2.0

PLANT NUMBERING PROCEDURE

CHAPTER No	TITLE	VALID REVISION
1	Machinery and Piping – (Based on Old PS-16)	1
2	Electrical Numbering – (Based on Old PS-10)	1
3	Instrument Numbering – (Based on Old PS-14)	1
4	Marking and Nameplates for Instrumentation and Electrification	

PROJECT STANDARD NO: 21

Chapter 1 - Machinery and Piping

PLANT NUMBERING PROCEDURE

Chapter 1 - Machinery and Piping

CONTENTS

- 1 Preface.
- 2 Annotation.
- 3 Short Introduction.
- 4 Plant Areas.
- 5 Numbering Machinery and Tanks.
- 6 Numbering Piping.
- 7 Numbering Manual Valves.

APPENDIX

- I Classification and allocation of sequential numbers for machinery and tanks.
- II Allocation of sequential numbers for manual valves.
- III Sub-process codes for piping.
- IV Flow substance codes.
- V Pipe labelling colour code.

1 PREFACE

1.1 Foreword

This chapter forms part of the

PLANT NUMBERING PROCEDURE

Manual issued in the form of separate chapters each dealing with a specific field.

Any deviation from this standard is always subject to consultation with the mills engineering lead and require a permission in writing.

This standard is subject to periodic review. It is revised frequently to meet the changing conditions and technical progress. Users are advised to secure the latest edition.

When using this publication, it is important to consider the context in which a particular paragraph or requirement is located. It is recommended that care be used to not lift a given requirement out of context.

1.2 This Chapter

This chapter describes the Mechanical numbering procedure.

1.3 Objective

The objective of this document is:

- to give guidance for the formulation and application of the mechanical identifiers to machinery and piping.
- to provide the background information necessary to understand the system as a whole.

2 ANNOTATION

The numbers will be allocated by Shotton Mill Ltd, which may, if practical, reserve certain numbers for specific purposes by others.

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping

3 SHORT INTRODUCTION

The plant numbering (hereinafter referred to as position numbers) forms the basis for the mill planning, administration, supervision, erection and operation. The position numbers will appear in drawings, flow sheets, reports, equipment lists and databases, manuals etc. The assets register is based on these numbers and they or their derivatives are also used as a filing address for drawings.

Basically all numbering is strictly hierarchical.

3.1 Naming Machinery

When giving linguistic names to the machinery, the limitations in the number of available characters in the control system shall be borne in mind not to cause unnecessary hassle at a later date, e.g. in the form of strange abbreviations.

4 PLANT AREAS

The complete plant is subdivided into a number of areas (process departments) mainly based on their physical function in the production chain. These plant areas are:

<u>Plant Area</u>	<u>Description</u>
10	Wood Handling
20	TMP Plant
30	Paper Mill PM1
35	Paper Mill PM3
36	Starch Process and Storage
40	Boiler Plant
42	Boiler Plant 7
45	Combined Heat and Power
50	RCF 1 (Recycled Fibre Plant 1)

52	RCF 1 (Recycled Fibre Plant 2)
53	RCF 3 (Recycled Fibre Plant 3)
55	OCC Warehouse & Fibre Plant
56	OCC
59	M.R.F. (Materials Recycling Facility)
60	Fresh and Effluent Water Treatment
65	E.T.P New Effluent Treatment Plant
70	Services and Auxiliary Departments
75	Site Services
80	Paper Mill PM2
85	Corrugator
91	Tissue machine TM1
92	Tissue machine TM2
93	Tissue Machine TM3

The plant area identifiers are used throughout the numbering procedure, whether mechanical, electrical, instrumentation or automation as a higher level designation to distinguish between the major parts of the plant.

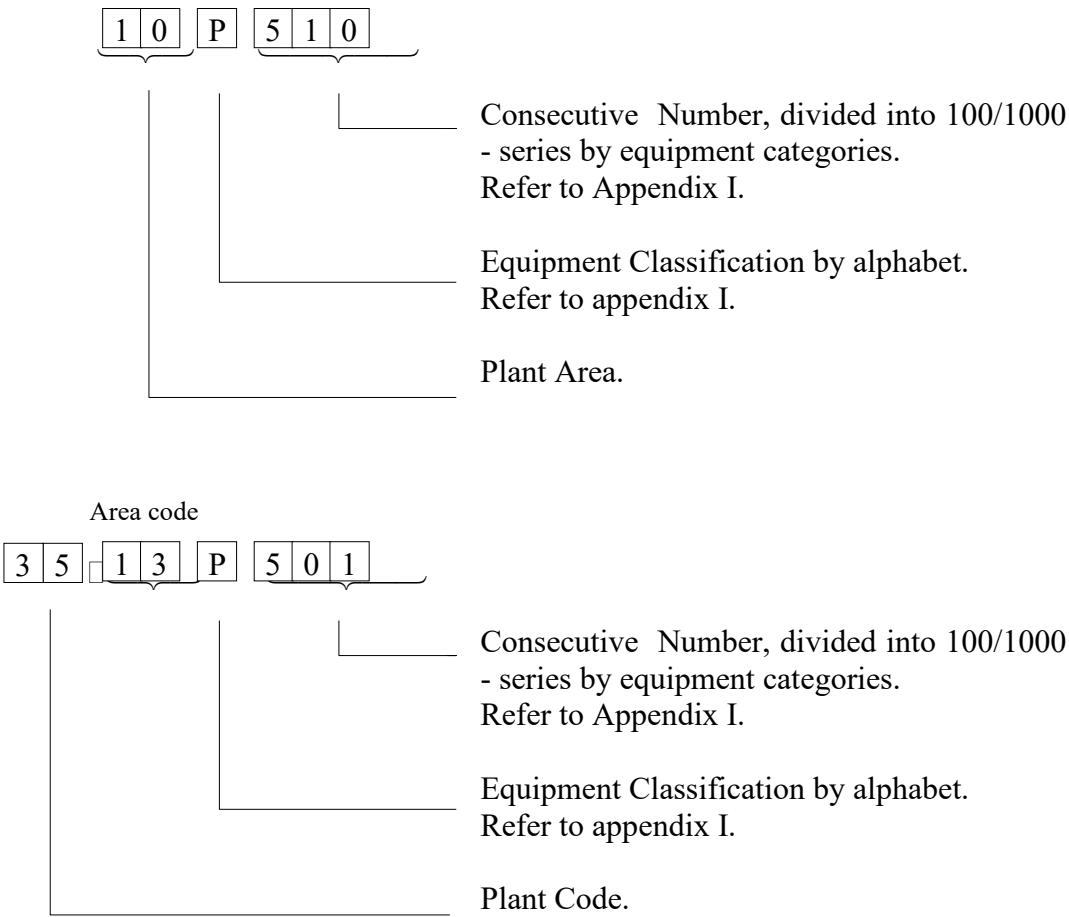
5 NUMBERING MACHINERY AND TANKS

5.1 Introduction

The mechanical position number (equipment designation, asset number) has a hierarchical structure and will additionally carry information of the kind equipment.

The use of the higher level designation is essential to fulfil the requirements for a unique identifier as the consecutive numbering of the identification is repetitive in other areas.

5.2 Structure



Old example:

- 10P510 = A pump in Area 10 (Wood Handling).
- 20P510 = A pump in Area 20 (TMP).
- 20V810 = A ventilation apparatus in Area 20 (TMP).

New example:

- 35-22P501 = A pump in Area 35 (PM3, approach flow system).
- 55-30P561 = A pump in Area 55 (OCC, pulping line 1).

5.2.1 Plant Area

Highest level designation. Refer to Section 5.

5.2.2 Equipment Classification

Letter identifiers are used to express the equipment classification by function. For applicable letters of the alphabet refer to Appendix I.

It shall be noted that the letters C, E, G, H and N are reserved for the purposes of electrical circuit numbering. Refer to the particular chapter for Electrical Numbering.

5.2.3 Consecutive Numbers

To further enhance the equipment classification the consecutive numbers are divided into 100/1000 - series and allocated to alphabetic identifiers.

The allocation of number series to letter codes is presented in Appendix I.

PROJECT STANDARD NO: 21

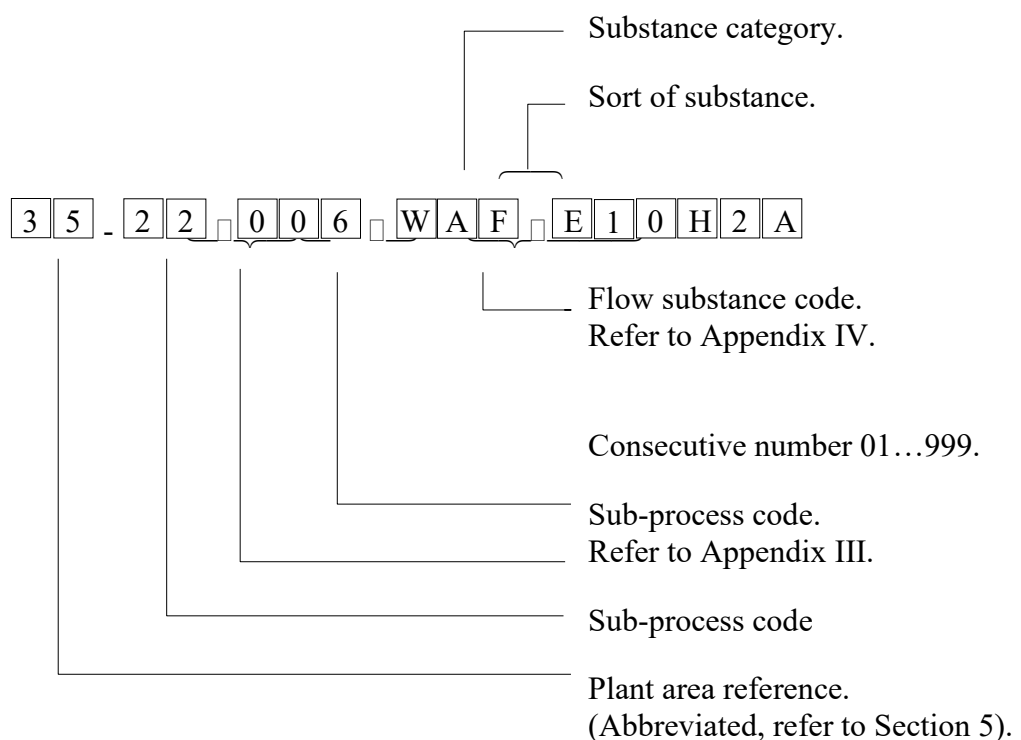
6 NUMBERING PIPING

6.1 Introduction

The pipelines are numbered area-wise referring to the sub-process which they serve.

6.2 Structure

The pipeline designation is a two part structure separated by a dash, the first part consisting of a five digit numeric identifier and the second part of a three letter definition for the flow substance.



New example:

35-22-006-WAF-100-E10H2A = PM3, Approach flow system, White water piping, line No.06, flow substance being white water.

6.2.1 Plant Area Reference

The plant area identifier has been abbreviated into one-digit expression i.e. the designation is the same as the first digit of the plant area.

Refer to Section 5.

6.2.2 Sub-process Code

Refer to Appendix IV for complete Flow Substance Codes.

6.2.3 Consecutive Number

The consecutive numbers run for each designated sub-process starting 01. 999.

Should more than 999 numbers for piping a certain sub-process be required, the domain shall be extended by allocating another sub-process code to the subsystem in concern. (+ 4 digits if required)

6.2.4 Flow Substance Code

Flow substance code is a hierarchical alphabetic combination where the first digit is designed to give information on the category of a particular substance and the following two letters are designed to specify the sort of substance.

The flow substance categories (first letter) are:

<u>Category</u>	<u>Substance</u>
A	Air
B	Additives, Chemicals
C	Condensate
E	Effluent
F	Feed Water
G	Gas
O	Oil
P	Pulp
S	Steam
W	Water

6.2.5 Pipe Labelling Colour Code

Pipe identification labels shall be colour coded as shown in Appendix V.

7 NUMBERING MANUAL VALVES

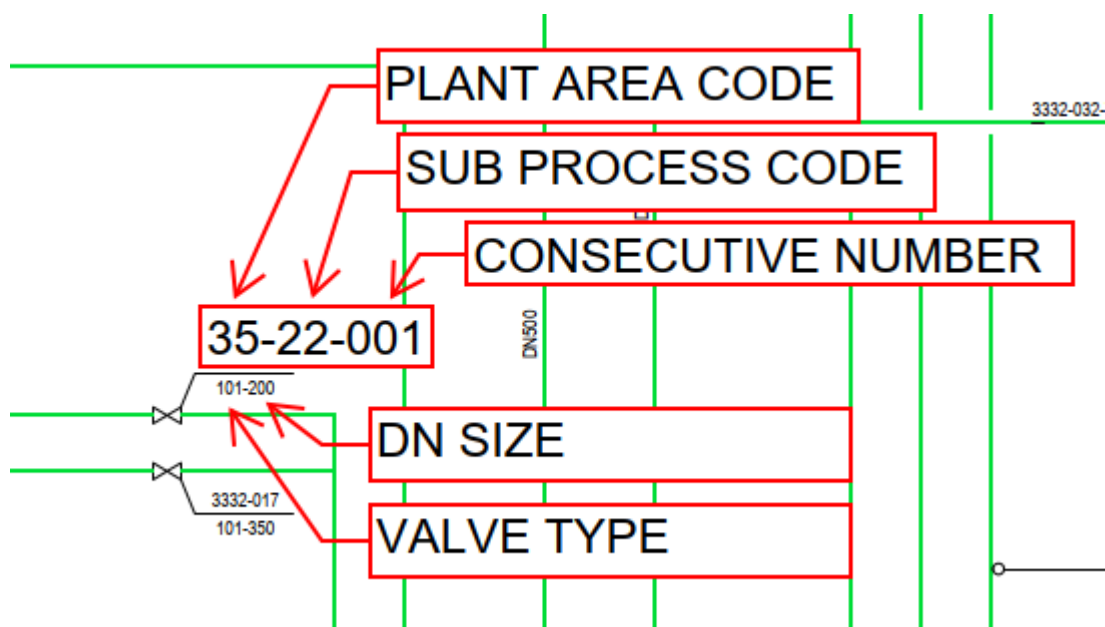
7.1 Introduction

The manual shut-off valves, check valves, steam traps, flow indicators etc., which are not equipped with an actuator for remote control, shall be baptised as described in this section.

Numbering procedure for control valves differs to this one and is described separately in the numbering procedure for instrumentation.

7.2 Structure

Presentation as in P & I diagram.



7.2.1 Valve Number

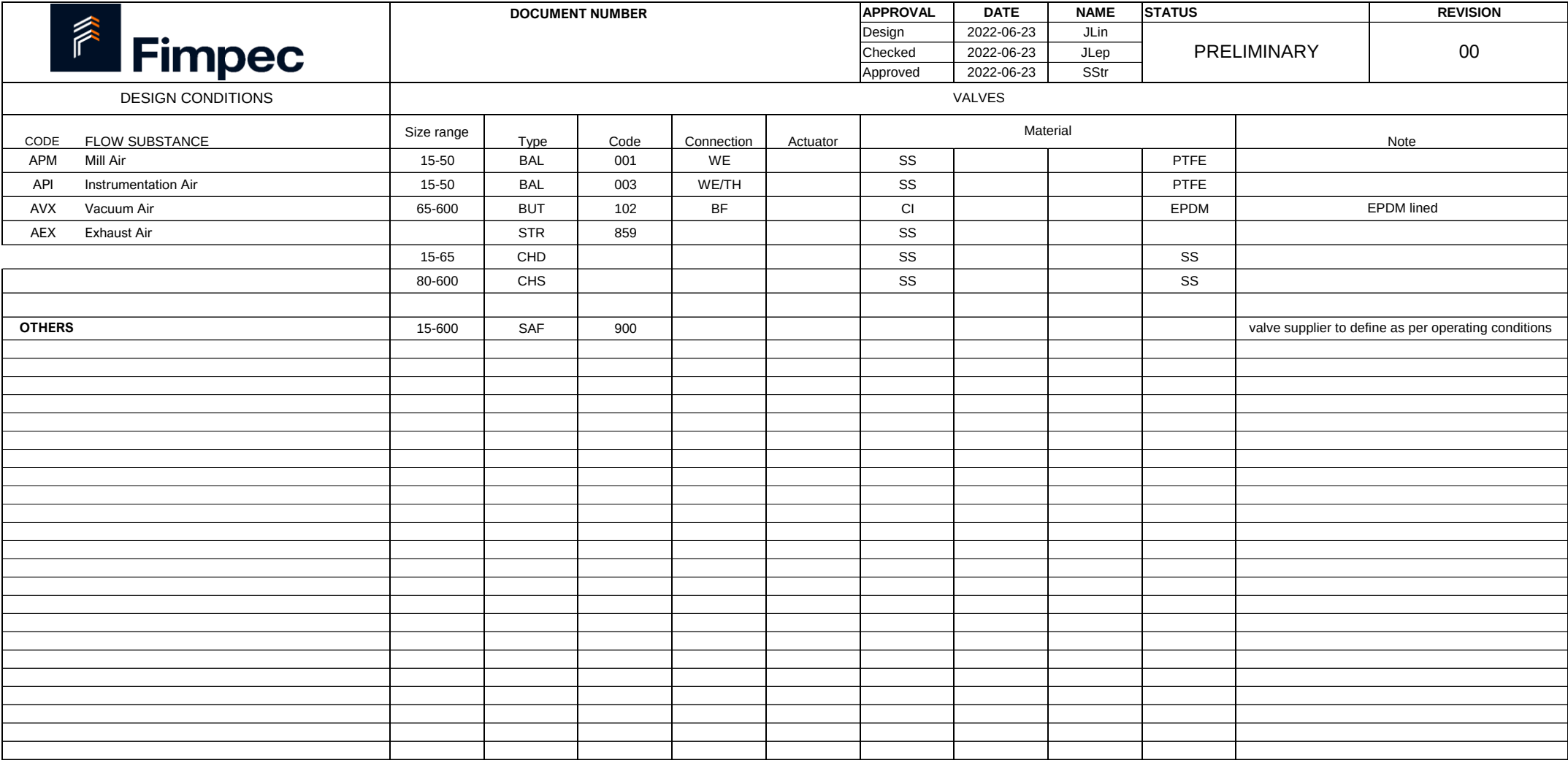
As above example consecutive numbers.

7.2.2 Valve Data

Valve Type List (FIMPEC – Rev 00)

	DOCUMENT NUMBER						APPROVAL	DATE	NAME	STATUS	REVISION	
							Design	2022-06-23	JLin	PRELIMINARY		00
							Checked	2022-06-23	JLep			
							Approved	2022-06-23	SStr			
DESIGN CONDITIONS		VALVES										
CODE	FLOW SUBSTANCE	Size range DN	Type	Code	Connection	Actuator	Material				Note	
							Body	Disc	Spin	Seat		
WATERS, CONSISTENCY < 0,5 %												
WAB	Clear Filtrate	15-100	BAL	001	WE		SS			PTFE		
WAA	Cloudy Filtrate	15-100	BAL	003	WE/TH		SS			PTFE		
WAC	Super Clear Filtrate	15-100	BAL	021	FL		SS			PTFE		
WFC	Cooling Water	125-700	BUT	101	BF		SS			SS		
WM	Fresh Water	80-600	CHS	600	BF		SS			SS		
WHC	Circulation Water for Heating	15-65	CHD	640	BF		SS			SS		
WSH	Shower Water		STR	854	FL		SS			PTFE	Air trap	
WFL	Sealing Water		STRA	866	FL		SS			PTFE	Mud pan	
WWL/M/H	Warm Water LP/MP/HP		STR	880	TH		SS			PTFE	Air trap	
WAF	White Water	15-100	FBV	885	WE		SS			PTFE	Flow balancing vave	
		125-300	FBV	886	FL		SS			PTFE	Flow balancing vave	
WATERS, CONSISTENCY > 0,5 %												
EFC	Fibre Containing Effluent	25-250	BAL	030	WE		SS			STELL		
ELX	Effluent Sludge	100-600	SLI	355	BF		SS					
STOCK & BROKE, CONSISTENCY > 0,5 %												
PP	Broke	15-100	BAL	001	WE		SS			PTFE		
LFS	Long Fiber Stock (From OCC)	15-100	BAL	003	WE/TH		SS			PTFE		
PS	Paper Stock	15-100	BAL	021	FL		SS			PTFE		
PSR	Paper Stock Reject	125-700	BUT	101	BF		SS			SS		
SFS	Short Fiber Stock (From OCC)	100-600	SLI	350	BF		SS			EPDM		
OS	OCC Stock											
OSR	OCC Stock Reject											
STEAM												
LP	Low Pressure Steam	10-65	GLO	250	WE	HW	CS			STELL		
MP	Medium Pressure Steam	10-65	GLO	251	FL	HW	CS			STELL		
		80-800	GAT	331	WE	HW	CS			STELL		
		15-100	BAL	081	FL	HL	CS			PTFE	Carbon reinforced / Cylinder shut-off valve	

		DOCUMENT NUMBER					APPROVAL	DATE	NAME	STATUS	REVISION
							Design	2022-06-23	JLin		
							Checked	2022-06-23	JLep		
							Approved	2022-06-23	SStr		
DESIGN CONDITIONS		VALVES									
CODE	FLOW SUBSTANCE	Size range	Type	Code	Connection	Actuator	Material				Note
		15	NEE	522	TH	HW	CS			SS	
		15-100	CHD	640	BF		SS			SS	
		125-800	CHS	641	BF		SS			SS	Double swing
			SAF	800	FL		SS				Safety relief valve
			SAF	801	FL		CS				Safety relief valve
			SAF	802	FL		SS			SS	Vacuum relief valve
			SAF	810	FL		CS				Self-acting pressure reducing valve
			STR	853	FL		CS				Thermal air trap
			STR	859	FL		SS				Thermal air trap
HIGH TEMPERATURE WATERS											
CL	Live Steam condensate	15-65	GLO	250	WE	HW	CS			STELL	
CM	Live Steam condensate	15-65	GLO	251	FL	HW	CS			STELL	
*CLV	Condensate with Steam	15-100	BAL	081	FL	HL	CS			PTFE	Carbon reinforced / Cylinder shut-off valve
		15-100	BAL	001	WE	HL	SS			PTFE	Carbon reinforced / For SS pipes
		125-500	BUT	101	BF	HW/GE	SS			SS	For SS pipes
		80-500	GAT	331	WE	HW	CS			STELL	
		80-500	GAT	302	FL	HW	SS			STELL	
		15-100	CHD	640	BF		SS			SS	
		125-500	CHS	641	BF		SS			SS	Double swing
			SAF	812	FL		CS				Self-acting pressure limiting valve
			STR	856	FL		CS				Steam trap, float
			STR	857	FL		CS				Steam trap, thermal
			STR	854	FL		SS				Air trap, float
			STR	880	TH		SS				Air trap, float
			STR	859	FL		SS				Air trap, thermal
			STRA	860	FL		SS				Strainer
			STRA	865	FL		CS				Strainer
		15-100	FBV	885	WE	HL	SS			PTFE	Flow balancing valve
		125-300	FBV	886	FL	HW&GE	SS			PTFE	Flow balancing valve
NON-FLAMMABLE GASES											



SHOTTON

NOTE: SEE APPENDIX 1. DRAWING OF VALVES AROUND TANKS/TOWERS.

- Valve types shall be tested and found reliable in practice.
- Actuator for ball and butterfly valves shall be selected as follows:
 - lever DN ≤ 150
 - gear DN ≥ 200
- A slide (stock) valve (also known as knife gate valve) shall be used in the pump suction side and in the
- Ball valves will be used when valve size is DN100 or smaller. Ball valves must be full opening.

NUMBER IN DRAWING	PURPOSE	TYPE	VALVE CODE	DN
1.1	DRAIN ON TOWERS AND CHESTS/ TANKS	Slide	350	150/200
1.3		Butterfly	101	200
1.4		Butterfly	101	≥125
		Ball	001	≤ 100
2.1	DRAIN ON PUMP INLET	Ball	001	80
2.2		Ball	003	50
3.1	SHUT-OFF ON PUMP INLET	Slide	350	≥125
3.3		Butterfly	101	≥125
3.4		Ball	001	≤ 100
4.1	SHUT-OFF ON LINE	Slide	350	≥125
4.2		Butterfly	101	≥125
4.3		Ball	001	≤ 100
5	FLUSHING	Ball	003	50
6	LOCAL PRESSURE MEASUREMENT	Ball	003	15
7.1	SAMPLE	Ball	022	25
7.2		Ball	022	40
7.3		"Special"		
8.1	CHECK VALVE	Non-return swing	600	≥80
8.2		Non-return disc	640	≤ 65

tank drain connections for pulps, consistency over 3 %, when valve size is DN125 or bigger.

CONNECTION	CONSISTENCY	NOTE
BF BF BF WE	$\geq 3\%$	DN200 on towers, DN150 on chests Water tower Water tanks. Normally DN150 Special cases: Small tanks
WE WE/TH	$> 1,5\%$ $< 1,5\%$	for flushing
BF BF WE	$\geq 3\%$	
BF BF WE	$\geq 3\%$	Special cases only
WE/TH		
WE/TH		
WE/TH WE/TH WE	$< 3\%$ $> 3\%$	"Special" sample valves for LC-stock. Push-button valves with automation position
BF BF		

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping

Appendix 1

CLASSIFICATION AND ALLOCATION OF SEQUENTIAL NUMBERS FOR MACHINERY AND TANKS

L - Classification And Numbering-Series:

Process equipment	L001 - L299 continued L1001 - L2999
Hydraulics	L301 - L399 continued L3001 - L3999
Agitators	L401 - L499 continued L4001 - L4999

P - Classification And Numbering-Series:

Pumps	P501 - P599 continued P5001 - P5999
-------	---

T - Classification And Numbering-Series:

Tanks, Towers	T601 - T699 continued T6001 - T6999
---------------	---

H - Classification And Numbering-Series:

Cranes, Lifts, Electric doors	H701 - H799 continued H7001 - H7999
-------------------------------	---

V - Classification And Numbering-Series:

Ventilation	V801 - V999 continued V8001 - V9999
-------------	---

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping
Appendix 1

M - Classification And Numbering-Series:

Mobile plant, Vehicles ¹⁾ 70M001 - 70M999

S - Classification And Numbering-Series:

Slings, Lifting equipment (other than cranes) ¹⁾ 70S001 - 70S999

Z - Classification And Numbering-Series:

Ladders and Mobile access equipment ¹⁾ 70Z001 - 70Z999

¹⁾ For M, S and Z consecutive numbers are used throughout the mill site.

NOTE:

- A) RCF2 (area 52) and RCF3 (area 53) 1000-series numbering only.
- B) Numbers 300/3000, 400/4000, 500/5000, 600/6000, 700/7000 and 800/8000 are not used.

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping

Appendix II

SUB-PROCESS CODE For PIPING

In this appendix the sub-process codes are defined for pipe line, process equipment, electrification and automation numbering. Please note that there are variations in codes between the plant areas (departments).

ALLOCATION OF SEQUENTIAL NUMBERS FOR MANUAL VALVES

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping

Appendix IV

LETTER CODES FOR FLOW SUBSTANCES:

Air

Instrument and Ctrl Air	API
Mill Compressed Air	APM
Exhaust Air	AEX
Vacuum	AVX

Chemicals

Aluminium Sulphate	BAS	$\text{Al}_2(\text{SO}_4)_3$
Defoamer	BDF	
Surface Size Starch	BSS	
Sizing Agent	BSA	
Dyestuffs	BDS	
Sequestering DTPA-NTA	BRD	Na_3PO_4
Retention Agent	BRR	
Felt Washing Agent	BWL	
Wire conditioning agent	BCW	
Sodium Hydrosulphate	HS	
Talc Slurry	BCT	
Calcium Chloride	BCA	
Dispersing Agent	BFX	
Polymers (PAC)	BPX	
Soap Dilution	LTX	
Bentonite	BOL	
Hydrazine	DHY	N_2H_4
Morfoline	DMO	
Sodium Hydroxide	DNO	Caustic Soda, NaOH
Sodium Phosphate	DNP	
Sodium Silicate	DNW	Water Glass
Sodium Hypochloride	DNH	
Sulphuric Acid	DSA	H_2SO_4
Sulphuric Acid diluted	DSD	
Peroxide	DPX	
Broke	PP	
Paper Stock	PS	
Paper Stock Reject	PSR	
Silica	BSB	
Biocide	BIO	

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping
Appendix IV

Steam and Condensate

HP - Steam	SH
MP - Steam	SM
LP - Steam	SL
TMP Steam to Economizer	SEX
Condensate with Steam	CLV
Live Steam Condensate	CL
Live Steam Condensate	CM
Live Steam Condensate	CLP

Effluent

Fibre Containing Effluent	EFC	
Fibre Containing Indoors	EFC	
Effluent Sludge	ELX	
Storm Water Effluent	ERWI	indoors
Storm Water Effluent	ERWU	underground
Sanitary Effluent	ESW	

Gas

Gas Butane	GBU
Gas Propane	GPR
Gas Bio	GBI
Natural Gas	GNG
Liquid Gas	GLG
Hydrogen	GHG

Oil

Lube Oil Main Lines	OHL
Hydraulic Oil	OHY
Heavy Fuel Oil	OFH
Light Fuel Oil	OFL

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping
Appendix IV

Pulps

~~Chips~~

Chemical Pulp	PC
De-inked Stock	PD
De-inked Stock Reject	PDR
Ink Sludge	PDI
Broke	PP
Paper Stock	PS
Paper Stock Reject	PSR
OCC Stock	OS
OCC Reject stock	OSR
Long Fibre Stock	LFS
Short Fibre Stock	SFS

Waters

White Water	WAF	
White Water Clarified	WAC	
Cloudy Filtrate RCF	WAA	
Clear Filtrate RCF	WAB	
Make-Up Water	WTX	Feed Water System
Fresh Water	WM	
Fresh Water	WM	Underground
Cooling Water	WFC	
Sealing Water	WFL	
Potable Water	WFP	Drinking water
Potable Water	WFP	Underground
Warm Water HP	WWH	
Warm Water MP	WWM	
Warm Water LP	WWL	
Fire Fighting Water	WWF	
Fire Fighting Water for Sprinkler	WFS	
Circulation Water for Heating	WHC	
Boiler Feed Water HP	WBH	
Boiler Feed Water LP	WBL	
Shower water	WSH	

PROJECT STANDARD NO: 21

Chapter 1 – Machinery and Piping

Appendix V

CHAPTER 1 – MACHINERY PIPING LABEL COLOUR CODES

<u>PIPE CONTENT</u>	<u>LABEL COLOUR</u>	<u>COLOUR</u>	<u>IDENTIFIER</u>
FRESH WATER	DARK BLUE		WHITE LETTER
CLEAR FILTRATE	LIGHT GREEN		BLACK LETTER
CLOUDY FILTRATE	DARK GREEN		WHITE LETTER
R.C.F.	ORANGE		BLACK LETTER
BROKE	RED		BLACK LETTER
T.M.P.	ORANGE		BLACK LETTER
REC FIBRE	ORANGE		BLACK LETTER
INST AIR	WHITE		BLACK LETTER
MILL AIR	BLACK		WHITE LETTER
SEAL WATER	LIGHT BLUE		BLACK LETTER
SHOWER WATER	LIGHT GREEN		BLACK LETTER
V.H.P. WATER	DARK BLUE		WHITE LETTER
HIGH VAC	RED		BLACK LETTER
LOW VAC	YELLOW		BLACK LETTER
MEDIUM VAC	ORANGE		BLACK LETTER
MILL FIRE MAIN	RED		WHITE LETTER
STEAM	ORANGE		BLACK LETTER
COOLING WATER	DARK BLUE		YELLOW LETTER
CAUSTIC	YELLOW		BLACK LETTER
CONDENSATE	LIGHT GREEN		BLACK LETTER
WARM WATER	DARK BLUE		WHITE LETTER
WHITE WATER	DARK GREEN		WHITE LETTER
POTABLE WATER	DARK GREEN		WHITE LETTER
OXYGEN	BROWN		BLACK LETTER
CO ₂	BROWN		BLACK LETTER
ACIDS	DARK BLUE		WHITE LETTER
GAS	BROWN		BLACK LETTER
SLUDGE	PURPLE		WHITE LETTER

PLANT NUMBERING PROCEDURE

Chapter 2 - Electrical Numbering

CONTENTS

- 1 Preface.
- 2 Annotation.
- 3 Short Introduction.
- 4 Power Distribution Assemblies.
- 5 Control Gear Assemblies.
- 6 Circuit Numbering.
- 7 Item Designation.
- 8 Expression of Location.
- 9 Numbering of MCC and Rack rooms.

APPENDIX

- VI Definitions.
- VII Examples on Numbering.
- VIII Letter Codes for Equipment and Devices in Diagrams.

1 PREFACE

1.1 Foreword

This chapter forms part of the

PLANT NUMBERING PROCEDURE

Manual issued in the form of separate chapters each dealing with a specific field.

Any deviation from this standard is always subject to consultation with the mills engineering lead and requires permission in writing.

This standard is subject to periodic review. It is revised frequently to meet the changing conditions and technical progress. Users are required to secure the latest edition.

When using this publication, it is important to consider the context in which a particular paragraph or requirement is located. It is essential that care be used to not lift a given requirement out of context.

1.2 This Chapter

This chapter describes the electrical numbering procedure.

This chapter is based on guidelines set by IEC 750 - Item Designation in Electrotechnology.

1.3 Objective

The objective of this document is:

- to give guidance for the formulation and application of the electrical identifiers.
- to provide the background information necessary to understand the system as a whole.

2 ANNOTATION

The numbers will be allocated by Shotton Mill Ltd, which may, if practical, reserve certain numbers for specific purposes by others.

3 SHORT INTRODUCTION

The approach to electrical numbering is based on two different needs arising on the one hand from the distribution type electrical equipment requirements serving several customers (e.g. plant power distribution) and on the other hand from the individual circuit design (e.g. a motor circuit).

The purpose built electrical/rack room and hydraulic room numbering is an independent procedure.

Definitions for the purpose of this standard are presented in Appendix VI.

For plant areas refer to Table 1 below:

PLANT AREA	DESCRIPTION
10	Wood Handling
20	TMP Plant
30	Paper Mill PM1
35	Paper Mill PM3
36	Chemical Building
40	Boiler Plant
42	Bolier Plant 7
47	Low Pressure Condensing Turbine including Pump House and Dual Media Filter
50	RCF 1 (Recycled Fibre Plant 1)
52	RCF 1 (Recycled Fibre Plant 2)
53	RCF 3 (Recycled Fibre Plant 3)
55	OCC Warehose & Fibre Plant
56	OCC

59	M.R.F. (Materials Recycling Facility)
60	Fresh and Effluent Water Treatment
65	E.T.P (New Effluent Treatment plant)
70	Services and Auxiliary Departments
75	Site Services
80	Paper Mill PM2
85	Corrugator
91	Tissue machine TM1
92	Tissue machine TM2
93	Tissue Machine TM3

4 POWER DISTRIBUTION ASSEMBLIES

For examples refer to Appendix VII.

PROJECT STANDARD NO: 21

Chapter 2 - Electrical Numbering

4.1 Characteristic Letters and Consecutive Numbering

<u>Letter</u>	<u>Description</u>	<u>Numbering</u>
A	132 kV Substation	1 ... 9
B	Spare	n/a
C	11 kV Switchgear assembly, main distribution level	01 ... 99
D	Spare (6.6 or 3.3 kV)	01 ... 99
E	690 V MCC 690 V Sub-MCC	01 ... 99 101 ... 999
K	Lighting and Maintenance Distribution Lighting Sub-Boards Emergency Lighting Sub-boards	0001 ... 9999 1001 ... 9999 501 ... 599
H	Sectional Drive Distribution Sectional Drive Sub-distribution	01 ... 99 101 ... 999
R	Rectifier Panels	11 ... 99

4.2 Power Transformers

The transformers will be assigned with letter T combined with associated switchgear assembly identifier.

Example:

TC1	Transformer feeding 11 kV switchgear assembly No.1.
TC2	Transformer feeding 11 kV switchgear assembly No.2.
TE05	Transformer feeding 690 V MCC No.05.

4.3 Switchgear Assemblies

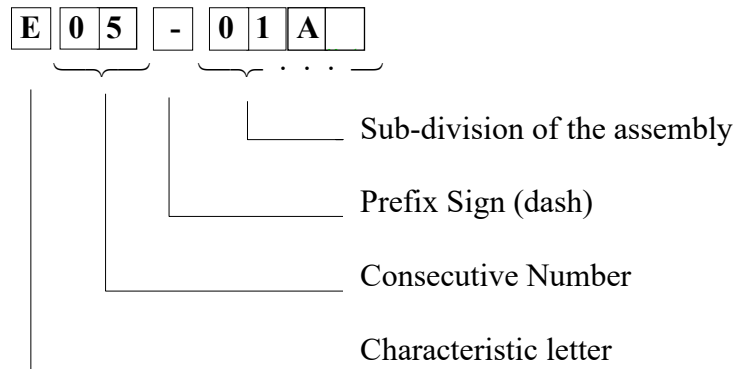
Characteristic letters by voltage/system reference followed by a consecutive number are assigned to distribution equipment.

A switchgear or control gear assembly as a whole is further sub-divided on the grid location principle.

PROJECT STANDARD NO: 21

Chapter 2 - Electrical Numbering

4.4 Structure



Identifiers for subassemblies are twofold:

- for Medium Voltage switchgear (at Shotton 11kv) they are derived from the main assembly identifier by adding a consecutive number to board number starting 1...for each sub-system.
- for Low Voltage switchgear (at Shotton 400v or 690v) 100 - series consecutive numbers are used.

Example:

C2	=	11 kV Medium Voltage main switchgear assembly No.2.
C21	=	11 kV Medium Voltage subassembly No.1 fed from the main board C2.
C21-5	=	Switchgear field No.5 in the 11 kV subassembly C21, fed from the main board C2.
E05	=	690 V MCC No.05.
E05-01C	=	A starter in location C in field 01 in 660 V MCC No.05.
E108	=	690 V Sub-MCC (NOTE: no direct reference to feeding board).
K16	=	400 V Lighting and Maintenance main assembly No.16.
K108	=	400 V Lighting sub-panel No.108 (NOTE: no direct reference to feeding board).

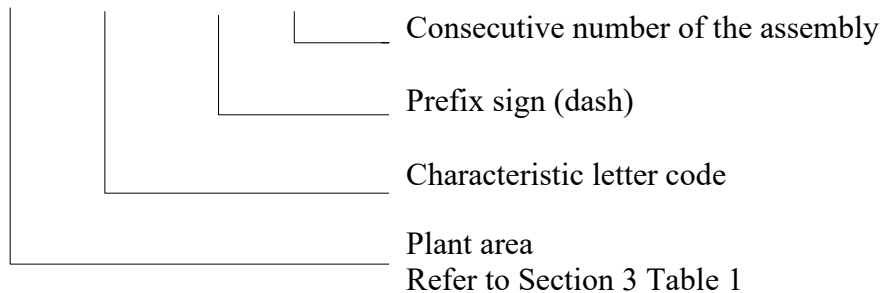
5 CONTROLGEAR ASSEMBLIES

Basically any common housing such as control panels, relay racks, instrumentation junction boxes, system cabinets and racks, I/O cabinets and racks, voltage distribution panels, PC keyboards, etc. will come under this procedure. Typically components of more than one circuit/loop/system are installed in the above described enclosures or cabinets.

5.1 Structure

Old example:

2 0 C R - 2 1



New example:

35-69CR-021 = Plant area 35 (PM3), sub area 69 (MCS/DCS) control rack assembly No.021 (consisting of several individual cabinets).

35-57CB-005 = Plant area 35 (PM3), sub area 57 (Press section) control box assembly No.005.

Old example:

20CR-21 = Area 20 (TMP) control rack assembly No.21 (consisting of several individual cabinets).

Control gear assemblies are sub-divided like the power distribution assemblies on the grid location principle. However, on individual cabinets they are separated by a point and an additional number.

New Example:

20CR-21.04 = The cabinet No.4 in the control gear assembly 20CR-21.

PROJECT STANDARD NO: 21

Chapter 2 - Electrical Numbering

If the cabinets are two sided and there is a need to distinguish between the halves, letters

- F for the front side.
- B for the back side.

shall be used as a suffix to the cabinet identifier.

New example:

35-69CR-021.04F = Plant area 35 (PM3), sub area 69 (MCS/DCS) control rack assembly No.021 (Front side of the cabinet No 4).

5.2 Characteristic Letter Codes for Equipment and Enclosures and Numbering

More letter codes for MCS will be added later.

<u>Letter Code</u>	<u>Description</u>	<u>Numbering</u>
CB	Control Box	001 ... 999
CP	Control Panel, Operator Terminal etc.	001 ... 999
VD	Control Voltage Distribution Panel	001 ... 999
RR	Relay Rack	001 ... 999
PC	Input/Output Rack	001 ... 999
VS	Video Screen/Keyboard	001 ... 999
CR	Cross Connection Rack, Control Rack (including. DCS I/O)	001 ... 999
PR	Printer	001 ... 999
MC	Micro Computer	001 ... 999
HP	Hydraulic Panel	001 ... 999
JB	Junction Box (typically for one circuit)	001 ... 999

JB	Instrument Junction Box	001 ... 999
SVB	Solenoid Valve Box	001 ... 999

Revised Information (08.04.2022)

001 – 499	Equipment related to MCS.
500 – 999	Equipment related to DCS.

6 CIRCUIT NUMBERING

The number of an electrical circuit is basically determined by its function and whether it is associated with machinery or other equipment or utility service. The numbering is divided into:

- 1) Circuits associated with machinery.
- 2) Miscellaneous circuits.
- 3) Power distribution.

6.1 Circuits Associated with Machinery

When a circuit is closely associated with a machine or equipment assigned with the mechanical position number, the electrical circuit number is its derivative. To obtain the circuit number, the letter for mechanical equipment classification is substituted for a dash and a sequential number separated by a dot, is added to it as a suffix.

Old example:

Mechanical position:	20P510
	↓
Associated general circuit:	20-510.0
Associated 1 st motor circuit:	20-510.1
Associated 2 nd motor circuit:	20-510.2

New example:

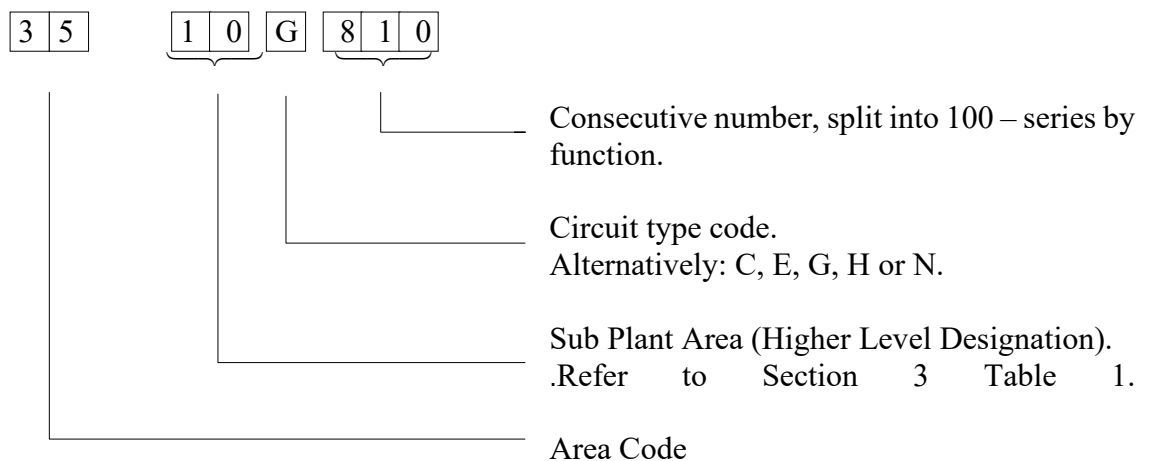
Mechanical position:	35-22P501
	↓
Associated general circuit:	35-22-501.0
Associated 1 st motor circuit:	35-22-501.1
Associated 2 nd motor circuit:	35-22-501.2

- Suffix 0: The number zero (0) is used as a suffix for control type circuits not including a motor.
- Suffixes 1: The numbers 1 are used as suffix for motor circuits; a number per motor, e.g. a disc filter can have four (4) motors, i.e. four associated circuits, but being one machine with one mechanical position number only.

6.2 Miscellaneous Circuits

Circuits not closely associated with any mechanical position i.e. there is no mechanical position number available, will be assigned with a circuit numbers through a separate procedure utilising similar structure to that of the mechanical position number.

6.2.1 Structure New Example



6.2.2 Circuit Type Code

Applicable codes:

<u>Letter</u>	<u>Description</u>
C	Computer Systems and Data Cabling.
E	Process related control type circuits.
G	Voltage Distribution, Group Starts, Cabling type circuits (including earthing circuits).
H	Motorised valves.
N	Electrical circuit to serve instrument loop (e.g. a motor for consistency transmitter).

6.2.3 Consecutive Number

Allocation of sequential number series:

<u>Sequential No.</u>	<u>Description</u>
0001 ... 0099	Non-Specific Circuits
1000 ...	
0800 ... 0899	Voltage Distribution
8000 ...	
0900 ... 0999	Earthing Circuits
9000 ...	

Example:

35-1G810

10G810 = A voltage distribution (VD) circuit in Area 10 (Wood Handling).

A number under which voltage supply cabling is engineered and installed from a dedicated VD panel. Another VD panel in the same area would use a different G-number.

Individual voltage supply cables would be called:

OLD	New
– 10G810-W11	35-10G810-W11
– 10G810-W12	35-10G810-W12
– etc.	

6.3 Power Distribution

The high and medium voltage circuit numbering is generally based on the idea of using the switchgear identifier also as a circuit identifier.

Example:

= C2-3 Would be a switchgear in field 3 in 11 kV assembly C2.

= C2-3 The designation for related circuits.

Low voltage distribution like e.g. incomers are considered as miscellaneous circuits.

However, alarm and other mixed auxiliary arrangements throughout all voltage levels will fall under the procedure designed for miscellaneous circuits.

7 ITEM DESIGNATION

7.1 Identification of Item

Each item shall be identified with an asset number. Numbers are used for distinguishing different items with the same letter code. Item designations are based on successive sub-divisions of a higher asset number. As described in Appendix VI section 1.3 and Appendix VIII.

An item designation is generally assigned when the circuit diagram is drawn.

NB Special numbers have been reserved and standardised for items such as cables, which are often similar in various circuits. The standardisation is explained in conjunction with the kind of item letter code table.

PROJECT STANDARD NO: 21

Chapter 2 - Electrical Numbering

Example:

OLD		NEW
20-510.1-M1	Motor 1, circuit 20-510.1.	35-20-510.1-M1
20-510.1-W1	Power cable 1, circuit 20-510.1.	35-20-510.1-W1
20-510.1-K1	Contactor 1, circuit 20-510.1.	35-20-510.1-K1
20-510.1-K1:13	Point of connection no. 13 of contactor K1 in circuit 20-510.1.	35-20-510.1-K1:13

7.2 Terminal Designation

Terminal designation shall correspond to the markings on the item, when terminals are not marked on the item a designation shall be assigned on the diagram (fictitious terminal designation).

7.2.1 Terminals in Main Current Circuits

<u>Marking</u>	<u>Description</u>	<u>Note</u>
L1	Phase 1	AC
L2	Phase 2	AC
L3	Phase 3	AC
N	Neutral	
L+	Positive	DC
L-	Negative	DC
PE	Protective Earth	
PEN	Earthed Neutral	
TE	Signal Earth	
E	Earth Conductor	

7.2.2 Terminals in Control Circuits

The terminals are considered to be and form an item for connections; designated as X. Thus the individual terminals are conductive parts of the item X and shall be designated with numbers. Item X can have one terminal or they can be arranged to a group (conventionally) forming a board or rail. Numbers are used for distinguishing different X-items.

For standardised X-rails refer to appendix VIII including earthing.

If several terminals to the neutral are needed they are connected in parallel by bars and assigned the same number.

Example:

Old	NEW
	35-20-510.1-x4:7
20-510.1-X4:7	Terminal 7 in terminal group/block/rail X4 (i/o rack), which is an item in circuit 20-510.1.
	NEW 35-20-510.1

7.2.3 Terminals in Earthing Circuits

The terminal bars for earthing terminations, signal earth (PE) or protective earth (PS), shall be designated with department number, letter X and a consecutive number. Example 42 x ø1.

NOTE: The letter PE or TE shall appear on the earthing bar labelling. See labelling standard in Part 4.

7.3 Position Specific Enclosures

If an enclosure is designed to house control or other equipment of only one circuit (position) it's understood as an item and its identification is determined respectively (refer to 7.1).

For defining the kind of code, the characteristic letter codes for equipment and enclosures as specified in 5.2 shall be applied. Numbers (1 ...) are used for distinguishing different items with the same kind of item letter code.

Example:

Old	New: 35-20-510.1-CB1
-----	----------------------

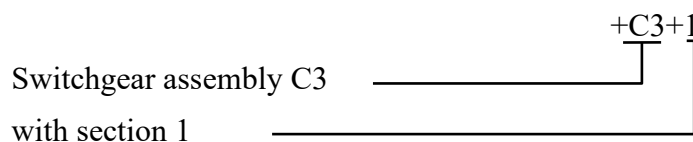
20-510.1-CB1	Control panel 1, housing only circuit 20-510.1 physical elements.
20-510.1-JB1	Junction box 1, circuit 20-510.1. New: 35-20-510.1-JB1
20-510.1-JB2	Junction box 2, circuit 20-510.1. New: 35-20-510.1-JB2

8 EXPRESSION OF LOCATION

Prefix sign plus (+) is used to express the physical location of an item.

Switchgear or control gear assemblies consist of a number of sections (cubicles). The assemblies are designated by letters, the sections by numbers.

Example:



Switchgear or control gear assemblies can be subdivided to give more detailed location. Vertical and horizontal mounting planes are provided with a grid system originating from the same point in each mounting plane and forming a module location, for example numbered vertically 01, 02, ... and horizontally A, B, C, ... (1, 2, ...). The location of an item is by reference to the point nearest to the origin of the grid system of the particular mounting plane.

Example:

690V MCC assembly E11 +E11+02+A-K1
with section 02 _____
with vertical subsection A (starter module) _____
with contactor K1 _____

Area 20 I/O rack assembly 21 35+20CR-21+17F+5.2-X4
with section 17 front side _____
with mounting plane reference _____
with X4 terminal block _____

If no confusion can arise, prefix signs may be omitted.

The combination with the item designation will provide supplementary information on the location of circuit components:

Item K1 of circuit 20-510.1 is located in MCC E11, in section 02, in module A

35=20-510.1-K1+E11+02+A

9 NUMBERING OF MCC AND RACK ROOMS

MCC and Rack Rooms will be numbered consecutively throughout the mill.

Where MCC and Rack Rooms are combined into one entity, the designation MCC is to be used.

9.1 MCC Rooms

New example:

35MCC01

Old example:

Room Reference MCC
Consecutive Number 1 ...

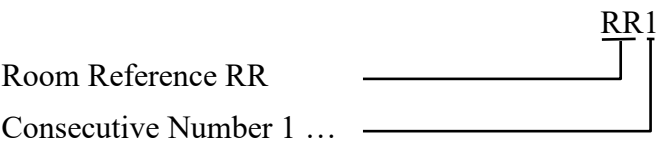
MCC1

9.2 Rack Rooms

New example:

35RR01

Old example:



9.3 Control Rooms

New example:

35CR01

PROJECT STANDARD NO: 21

Chapter 2 - Electrical Numbering

Appendix VI

DEFINITIONS

CONTENTS

1	INTRODUCTION.
1.1	Circuit.
1.2	Electrical Equipment.
1.3	Designation Block.
1.4	Prefix Signs.
1.5	Item.
1.6	Kind of Item.
1.7	Assembly.
1.8	Switchgear.
1.9	Switchboard.
1.10	Electrical Room.

PROJECT STANDARD NO: 21

Chapter 2 - Electrical Numbering

Appendix VI

1 INTRODUCTION

For the purpose of this standard, the following definitions shall apply.

1.1 Circuit

- 1) Administrative concept by means of which electrical engineering, maintenance and programming is managed.
- 2) An assembly of electrical equipment supplied from the same origin and protected against overcurrent by the same protective device(s).

NOTE: For Data, Voice and Fibre Optic cables see PSK Standard 5990.

1.2 Electrical Equipment

Any item for such purposes as generation, conversion, transmission, distribution or utilisation of electrical energy, such as machines, transformers, apparatus, measuring instruments, protective devices, wiring materials, accessories and appliances.

1.3 Designation Block

Part of a complete item designation (Asset number) with related information. In this standard four designation blocks are used.

- | | |
|----------|---|
| Block 1: | Higher level designation.
A plant area location – see section 3 table 1 |
| Block 2: | Location of item.
The physical location of an item in an assembly etc.. |
| Block 3: | Identification of item.
A distinctive code, which serves to identify an item in a diagram, list, chart and on the equipment. See Appendix III |
| Block 4: | Terminal designation.
Designation applied to the conducting parts of an apparatus which are provided for electrical connection to external circuits. |

PROJECT STANDARD NO: 21

Chapter 2 - Electrical Numbering

Appendix VI

1.4 Prefix Signs

The equal sign, plus sign, minus sign, and colon used to identify the various designation blocks.

The use of prefix signs enables the various designation blocks to be combined in a suitable and appropriate manner.

It shall be noted that if no confusion can arise the intermediate prefix sign(s) may be omitted. Generally this is possible only if the designation is based on the alternation of letter codes and numbers.

The prefix signs distinguish the various blocks (refer to 3.3):

Block	Prefix sign	Example
Higher level designation	=	=20
Sub Process Area		= 10
Location of item	+	+20CP101
Identification of item	-	-K1
Terminal designation	:	:01

1.5 Item

A basic part , component, equipment, functional unit, etc. which usually is represented by a graphical symbol on a diagram.

Example: Resistors, relays, generators, amplifiers, power supply units and switchgear assemblies may all be described as items for purpose of designation.

1.6 Kind of Item

Sort, variety, class or group of items regardless of their function in a circuit.

Example: Every type of resistor is regarded as being the same kind of item. A mechanical switching device may bear a different kind of item letter depending on whether it is used in a power circuit (on load switch = Q) or in a control circuit (control switch = S).

1.7 Assembly

A number of basic parts or subassemblies (two or more basic parts which form a portion of an assembly), or any combination thereof, joined together to perform a specific function, replaceable as a whole, but having a part or parts which are individually replaceable.

Example: Electric generator, audio frequency amplifier, power supply unit, switchgear assembly.

1.8 Switchgear

An assembly of main and auxiliary switching apparatus for operation, regulation, protection or other control of electrical installations.

1.9 Switchboard

An assembly of switchgear with or without instruments.

1.10 Electrical Room

Any room designed and built for permanent installation of electrical or control equipment is considered as electrical room.

Example: A room for a diesel driven emergency power generator is an electrical room.

Rack rooms are electrical rooms.

PROJECT STANDARD NO: 21

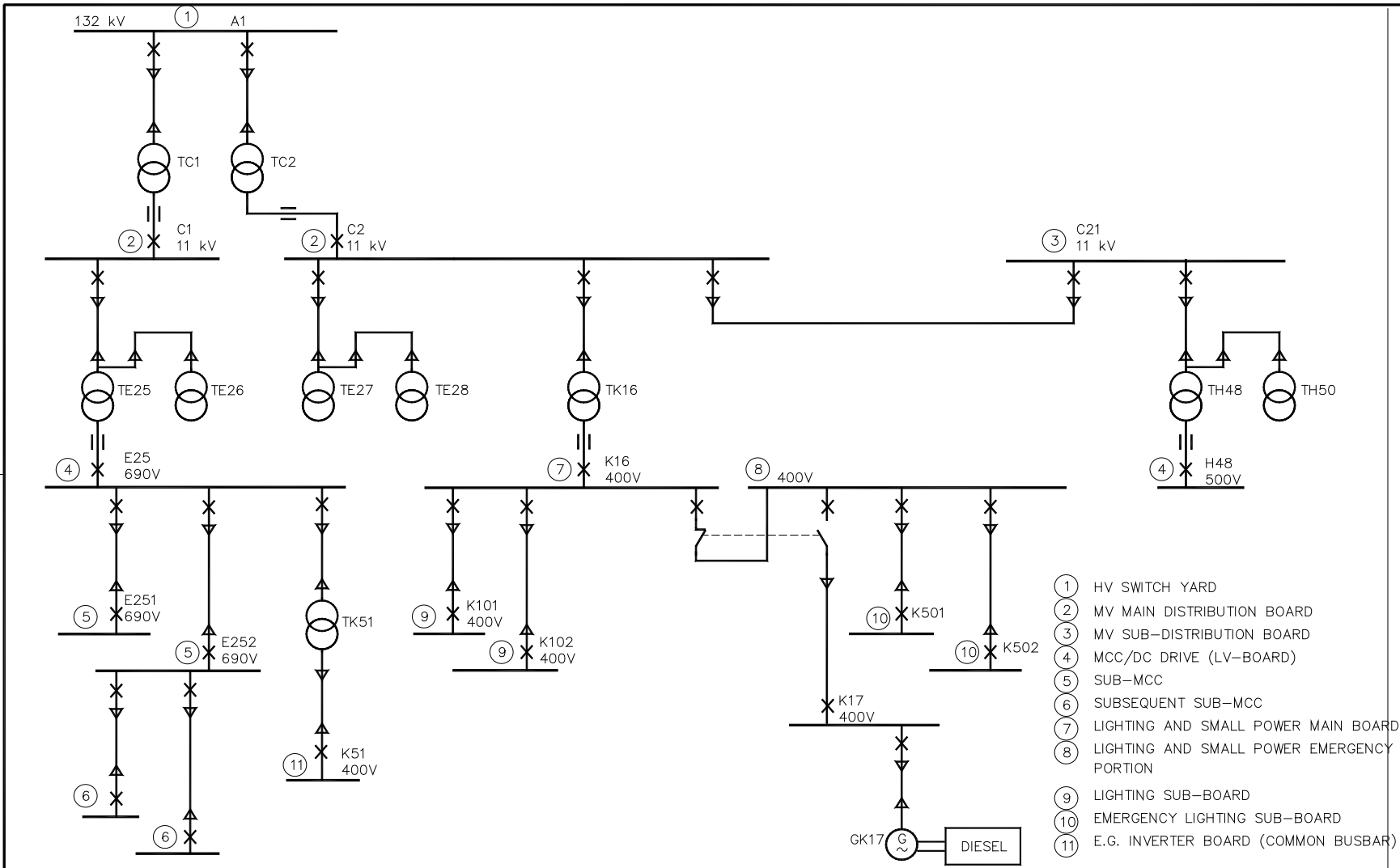
Chapter 2 - Electrical Numbering
Appendix VII

EXAMPLES ON NUMBERING

CONTENTS

PLANT NUMBERING PROCEDURE
CHAPTER 3 – ELECTRICAL NUMBERING

APPENDIX VII



JAAKKO PÖYRY

Drawn 27.10.99 pcs
Designed 27.10.99 JKS
Checked
Approved

Blinder Code
Archive Code



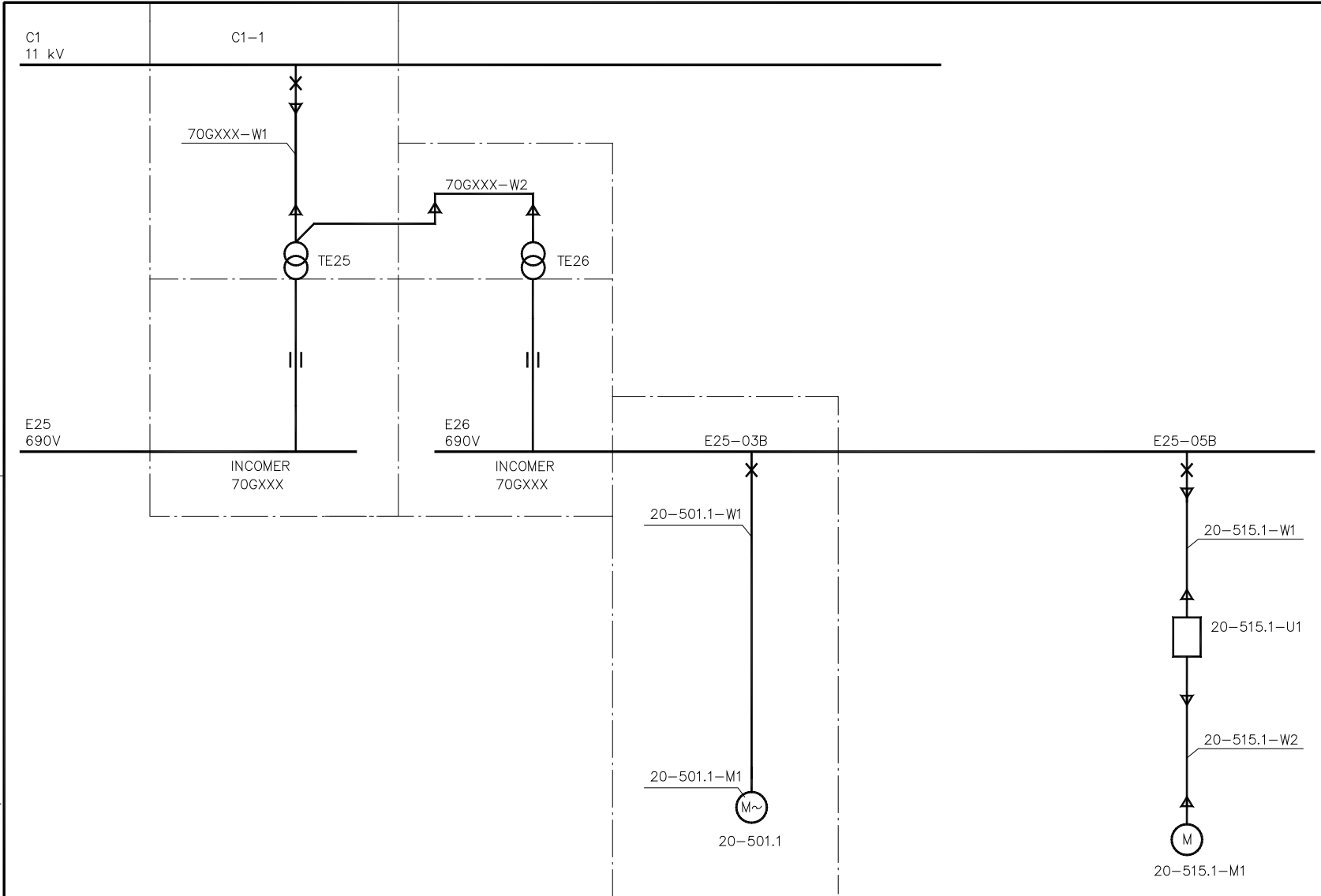
Department
Type E-STANDARD
Application EXAMPLE ON POWER DISTRIBUTION NUMBERING

JP No 16A1605
Dwg No
Sheet 2
Loop No

Supersedes Superseded

PLANT NUMBERING PROCEDURE
CHAPTER 3 – ELECTRICAL NUMBERING

APPENDIX VII



JAAKKO PÖYRY

Drawn 27.10.99 pcs
Designed 27.10.99 JKS
Checked
Approved

Binder Code
Archive Code
Department
Type E-STANDARD
Application EXAMPLE ON
CIRCUIT NUMBERING

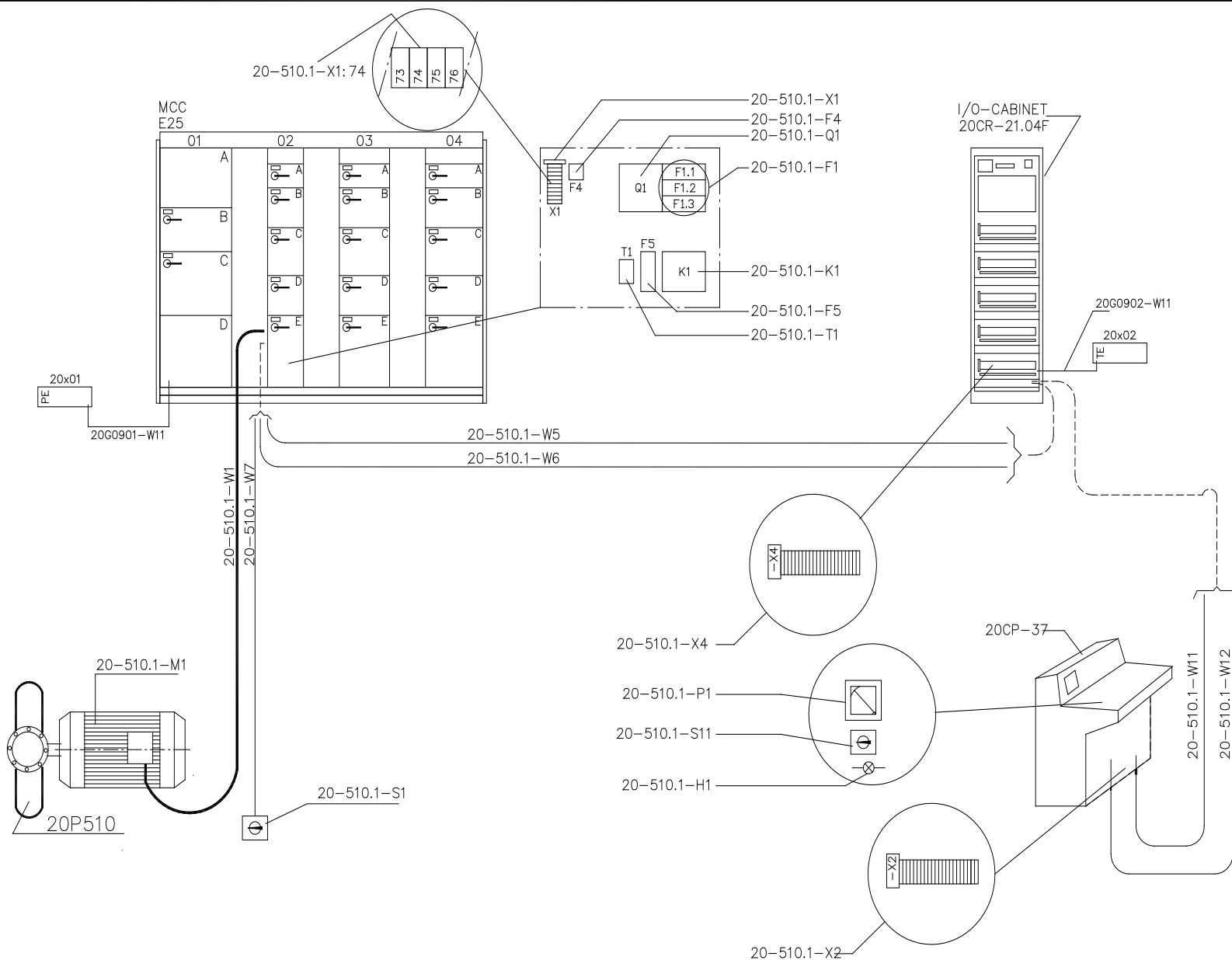
SHOTTON
UPM PAPER

JP No 16A1605
Dwg No
Sheet 3
Loop No

Supersedes
Superseded

PLANT NUMBERING PROCEDURE
CHAPTER 3 – ELECTRICAL NUMBERING

APPENDIX VII



PROJECT STANDARD NO: 21

Chapter 2 – Electrical Numbering

Appendix VIII

LETTER CODES FOR ELECTRICAL EQUIPMENT AND DEVICES IN DIAGRAMS

CONTENTS

PROJECT STANDARD NO: 21

Chapter 2 – Electrical Numbering
Appendix VIII

LETTER CODE	ELECTRICAL KIND OF ITEM	NUMBERING
A	Assemblies, subassemblies	1 ...
B	Transducers, from non-electrical to electrical quantity or vice versa	1 ...
C	Capacitors	1 ...
D	Binary elements, delay devices, storage devices	1 ...
E	Miscellaneous (not specified elsewhere)	1 ... Miscellaneous ES1 – Emergency Stop Switch
F	Protective devices	1... F1 Main fuse phase L1. F2 Main fuse phase L2. F3 Main fuse phase L3. F4 Control fuse (mcb) in a motor starter. F5 Thermal overload relay. F6 Ctrl fuse Circuit Breaker, Same as F4. F7 Spare. F8 Fuse for space heaters. F9 Spare. F10 Spare.
G	Generators, power supplies	1 ...
H	Signalling devices	1 ...
J	n/a	n/a
K	Relays, contactors	1 ...

Continued ...

PROJECT STANDARD NO: 21

Chapter 2 – Electrical Numbering
Appendix VIII

Continued ...

LETTER CODE	KIND OF ITEM	NUMBERING	
L	Inductors, reactors	1 ...	
M	Motors	1 ...	
N	Analogue elements	1 ...	
P	Measuring equipment, testing equipment	1 ...	
Q	Switching devices for power circuits, selectors	1 ...	Q1 Primary Switching Device e.g. in MCC.
			Q2 Secondary Switching Device.
R	Resistors	1 ...	
S	Switching devices for control circuits, selectors	1...	S1 Local switch in circuit (field switch).
			S7 Speed guard.
			S11 Panel Switch/Selector.
T	Transformers	1 ...	
U	Modulators, changers	1 ...	
V	Vacuum tubes, semiconductors	1 ...	
W	Transmission paths, waveguides, aerials, cables	11...	W1 Power cable.
			W2 Power cable.
			W3 Power cable.
			W4 Power cable.
			W5 MCC - Control Rack.
			W6 Current 0 ... 1A.
			W7 Local Switch S1.
			W8 Not yet specified.
			W9 Not yet specified.
			W10 Not yet specified.
			W11.. All other control cables.

Continued ...

Continued ...

LETTER CODE	DISCRIPTION		NUMBERING	
X	Terminals, plugs, sockets, earthing bars	1 ...	X1	Term. rail in Motor Starter.
			X2	Term. rail in field or in Sup- plier's boxes etc.
			X3	Term. rail in Control Panel.
			X4	Term. rail in I/O-rack.
Y	Electrically operated mechanical devices	1 ...		
Z	Terminations, hybrid transformers, filters, equalizers, limiters.	1 ...		

PLANT NUMBERING PROCEDURE

Chapter 3 - Instrument Numbering

CONTENTS

- 1 Preface.
- 2 Annotation.
- 3 Control gear Assemblies.
- 4 Loop Numbering.
- 5 Item Designation.

- | | | |
|----------|-----|--|
| APPENDIX | X | Plant areas and allocation of sequential numbers for instrument loops. |
| | XI | Principals and application of instrumentation functional codes. |
| | XII | Instrumentation symbols and presentation in P & I diagram. |

1 PREFACE

1.1 Foreword

This chapter forms part of the

PLANT NUMBERING PROCEDURE

-standard issued in the form of separate chapters each dealing with a specific field.

Any deviation from this standard is always subject to consultation with the mills engineering lead and requires permission in writing.

This standard is subject to periodic review. It is revised frequently to meet the changing conditions and technical progress. Users are required to secure the latest edition.

When using this publication, it is important to consider the context in which a particular paragraph or requirement is located. It is essential that care be used to not lift a given requirement out of context.

1.2 This Chapter

This Chapter describes the instrumentation numbering procedure.

This Chapter is based on guidelines set by PSK3601 – Industrial Process Measurement Control Functions and Instrumentation – Symbolic Presentation.

1.3 Objective

The objective of this document is:

- to give guidance for the formulation and application of the instrumentation identifiers and their symbolic presentation in P&I diagrams.
- to provide the background information necessary to understand the system as a whole.

2 ANNOTATION

The numbers will be allocated by Shotton Mill Ltd, which may, if practical, reserve certain numbers for specific purposes by others.

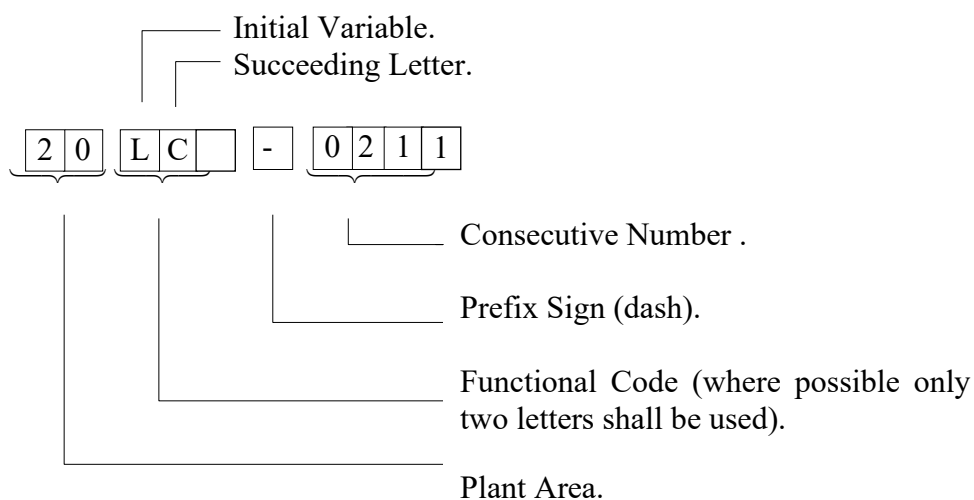
3 CONTROLGEAR ASSEMBLIES

In the field of electrical and instrumentation engineering, the numbering and identification procedure for control gear assemblies is common to both disciplines.

The procedure for control panels, relay racks, instrumentation junction boxes, system and I/O cabinet assemblies, etc. is described in the electrical part of the manual where reference shall be sought under the respective heading.

4 LOOP NUMBERING

4.1 Structure



Old example:

20LC-0211 = Level control loop no. 0211, area 20.

New example:

35-22LC-501 = Level control loop PM3, approach flow system.

4.1.1 Plant Area

For plant areas refer to Appendix 6.1, 6.2 and 6.3.

APPENDIX 6.1

Appendix 6.1 OCC Building Area Code 55

	OCC Building Plant Area Code 55
Sub-Process Code	Name
0	General
1	Rooms
2	Cranes, lifts and hoists
3	Compressed air
4	Instrument Air System
5	Safety showers
6	Fire and sprinkler water systems
7	CCTV
8	Building electrification
9	
10	Hall ventilation
11	Special room ventilation
12	Office building ventilation
13	Process ventilation
14	Steam and condensate
15	Heating water system (AHR)
16	Chilled water system
17	Sealing water system
18	Cooling water system
19	Potable water system
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	OCC pulping line 1
31	OCC pulping line 2
32	OCC cleaning and fractionation
33	OCC thickening
34	OCC reject handling
35	
36	
37	
38	
39	
40	
41	

42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	
67	
68	
69	MCS/DCS
70	Condition monitoring
71	
72	
73	
74	
75	
76	
77	
78	
79	
80	
81	
82	
83	
84	
85	
86	
87	Control system, separate PLC systems
88	
89	

90	
91	
92	
93	
94	
95	
96	
97	
98	
99	

Appendix 6.2
Paper Mill PM3 Plant Area Code

	Paper Mill PM3 Plant Area Code 35
Sub-Process Code	Name
0	General
1	Rooms
2	Cranes, lifts and hoists
3	Compressed air system
4	Instrument Air System
5	Safety showers
6	Fire and sprinkler water systems
7	CCTV
8	Building electrification
9	
10	Hall ventilation
11	Special room ventilation
12	Office building ventilation
13	Machine ventilation
14	
15	Heating water system (AHR)
16	Chilled water system
17	Sealing water system
18	Cooling water system
19	Potable water system
20	
21	Stock Preparation and Mixing
22	Approach Flow System
23	Vacuum system
24	Broke System
25	White water system
26	Warm- and shower water system
27	Steam and condensate system
28	Gas
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	
41	
42	
43	

44	
45	
46	
47	
48	
49	
50	
51	Surface size
52	Coating colour
53	Machine supply system(Machine circulation)
54	
55	Headbox
56	Forming section
57	Press section
58	Pre dryer section
59	
60	Sizing section
61	After dryer section
62	Calander
63	Coating section
64	Reel
65	
66	Hydraulics
67	Oil lubrication
68	Grease lubrication
69	MCS/DCS
70	Condition monitoring
71	QCS
72	Basis weight headbox
73	Steambox
74	Scanner 1
75	Scanner 2
76	
77	Moisturizer 1
78	Air Dryers
79	Infra Dryers
80	
81	WIS
82	Web break cameras
83	
84	Parent reel cart
85	Winder 1
86	Winder 2
87	
88	Roll handling system
89	Core handling system
90	Roll transport

91	Roll splitter
92	PM drives
93	Winder 1 drives
94	Winder 2 drives
95	
96	
97	
98	
99	

Appendix 6.3
CHP Building Plant Area Code

	CHP Building Plant Area Code 45
Sub-Process Code	Name
0	General
1	Rooms
2	Cranes, lifts and hoists
3	Compressed air
4	Instrument Air System
5	Safety showers
6	Fire and sprinkler water systems
7	CCTV
8	Building electrification
9	
10	Hall ventilation
11	Special room ventilation
12	Office building ventilation
13	Process ventilation
14	
15	Heating water system (AHR)
16	Chilled water system
17	Sealing water system
18	Cooling water system
19	Potable water system
20	
21	HRSG 1
22	HSRG 2
23	HSRG 3
24	Backup Boiler 1
25	Back up Boiler 2
26	Dearator and Feedwater System
27	Cooling Tower System
28	Balance of Plant
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	
41	

42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	Hydraulics
67	Oil lubrication
68	Grease lubrication
69	MCS/DCS
70	Condition monitoring
71	QCS
72	
73	
74	
75	
76	
77	
78	Air Dryers
79	
80	
81	
82	
83	
84	
85	
86	
87	
88	

89	
90	
91	
92	CHP Drives
93	
94	
95	
96	
97	
98	
99	

Appendix 6.4

Appendix 6.4 ETP Plant Area Code 65

ETP Building Plant Area Code 65	
Sub-Process Code	Name
0	General
1	Rooms
2	Cranes, lifts and hoists
3	Compressed air
4	Instrument Air System
5	Safety showers
6	Fire and sprinkler water systems
7	CCTV
8	Building electrification
9	
10	Hall ventilation
11	Special room ventilation
12	Office building ventilation
13	Process ventilation
14	
15	Heating water system (AHR)
16	Chilled water system
17	Sealing water system
18	Cooling water system
19	Potable water system
20	
21	
22	
23	
24	
25	
26	
27	
28	Natural Gas
29	
30	Primary treatment
31	Preacidification
32	Anaerobic treatment
33	Biogas handling
34	Flash aeration
35	Calcium separation
36	Activated sludge basin
37	Secondary clarification
38	Post cooling
39	Chemical handling
40	Sludge dewatering

41	
42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	Hydraulics
67	Oil lubrication
68	Grease lubrication
69	MCS/DCS
70	Condition monitoring
71	QCS
72	
73	
74	
75	
76	
77	
78	Air Dryers
79	
80	
81	
82	
83	
84	
85	
86	

ETP Building Plant Area Code 65	
Sub-Process Code	Name
0	General
1	Rooms
2	Cranes, lifts and hoists
3	Compressed air
4	Instrument Air System
5	Safety showers
6	Fire and sprinkler water systems
7	CCTV
8	Building electrification
9	
10	Hall ventilation
11	Special room ventilation
12	Office building ventilation
13	Process ventilation
14	
15	Heating water system (AHR)
16	Chilled water system
17	Sealing water system
18	Cooling water system
19	Potable water system
20	
21	
22	
23	
24	
25	
26	
27	
28	Natural Gas
29	
30	Primary treatment
31	Preacidification
32	Anaerobic treatment
33	Biogas handling
34	Flash aeration
35	Calcium separation
36	Activated sludge basin
37	Secondary clarification
38	Post cooling
39	Chemical handling
40	Sludge dewatering

87	
88	
89	
90	
91	
92	ETP Drives
93	
94	
95	
96	
97	
98	
99	

4.1.2 Functional Code

The use and application of letters describing the function of an instrument loop are discussed comprehensively in Appendix XI.

4.1.3 Consecutive Number

New Example, REVISED INFORMATION (23.3.2022)

001 – 499: Loops connected to MCS.

500 – 899: => 500 – 999 Loops connected to DCS.

9000 – 9249: Local loops, MCS.

9250 – 9499: Local loops, DCS.

OLD example: Basically the sequential numbers are dedicated to 0100/9000 – series:

0100 – series numbering: Loops connected to DCS.

9000 – series numbering: Local loops.

The sequential numbers are further broken down and certain series reserved for dedicated process areas.

For full allocation of numbers refer to Appendix X.

5 ITEM DESIGNATION

The item designation is divided into:

- 1) Device tagging.
- 2) Cable numbering.
- 3) Terminal numbering.

5.1 Device Tag

The device tag is derived from the instrument loop number by substituting the succeeding functional letter for instrument (device) letter.

For the instrument letters refer to Appendix XI Table 1.

The device description by letters is always two characters.

New example:

Level control loop no. 501, area 35, sub process area 22	35-22LC-501 ↓
Related control valve	35-22LV-501
Related level transmitter	35-22LT-501

Old example:

Level control loop no. 0135, area 20	20LC-0135 ↓
Related control valve	20LV-0135

Numbers are used for distinguishing different devices with the same kind of letter definition within a loop.

New example:

First control valve	35-22LV1-501
Second control valve	35-22LV2-501

Old example:

First control valve	20LV1-0235
Second control valve	20LV2-0235

5.2 Cable Numbering

All instrument cables, Fieldbus and Data, pneumatic and signal 4-20 mA, etc. between instruments, junction boxes, racks, panels and other devices shall be given a number based on the idea of destination in the field (e.g. transmitter, junction box). From the hierarchy point of view the lower level designations are referred to or if two systems the smaller one is used as a reference.

Cables are divided into two categories for the purpose of numbering:

- 1) Multicore cables.
- 2) Local cables.

5.2.1 Multicore cables

Multicore cables between the DCS system and junction box, PLC system, panels, etc. shall bear the number of the “satellite” accompanied by a consecutive number separated by a dash.

New example:

Instrument junction box 033, area 35, 35-22JBI-033-1
sub process code 22
with multicore cable no. 1

Consequently the 2nd multicore cable would be 35-22JBI-033-2 etc.

Control panel 221, area 35, 35-22CP-221-1
sub process code 22
with multicore cable no. 1

Old example:

Instrument junction box 33, area 20 20JBI-33-1
with multicore cable no. 1

Consequently the 2nd multicore cable would be 20JBI-33-2 etc.

Control panel 221, area 20 20CP-221-1
with multicore cable no. 1

5.2.2 Local cables

Local cables from field instruments and similar to junction boxes, PLC systems, panels, etc. shall bear the device tag accompanied by a consecutive number separated by a dash.

New example:

35-22LT-501-1

Control valve 1 of loop no. 501, area 35,

sub process code 22

with cable no. 1

Consequently the 2nd cable would be 35-22LT-501-2 etc.

Old example:

20LT-0235-1

Control valve 1 of loop no. 0235, area 20

with cable no. 1

Consequently the 2nd cable would be 20LT-0235-2 etc.

5.3 Numbering of Terminals

Electrical terminals in an instrument type junction box, panel etc. for use with multicore cable installation shall be numbered consecutively throughout starting 1 ...

Pneumatic terminals shall be numbered P1, P2, ...

Pneumatic distribution headers shall be named S1, S2, ...

The pneumatic header outlets shall be numbered S1/1, S1/2, ...

If required for clarity the electrical and pneumatic terminal rails may be distinguished by letters:

- P for pneumatic.
- L for electric.

**PLANT AREAS AND ALLOCATION OF SEQUENTIAL NUMBERS FOR INSTRUMENT
LOOPS**

CONTENTS

PROJECT STANDARD NO: 21

Chapter 3 – Instrument Numbering

Old system:

Plant areas and allocation of sequential numbers for instrument loops:

PLANT AREA	DESCRIPTION	DCS -LOOPS	LOCAL GAUGES
10	Wood Handling	0001 ... 8999	9000 ... 9999
20	TMP	0001 ... 8999	9000 ... 9999
30	PM1 Process Machine Finishing	0001 ... 2999 3000 ... 3999 7001 ... 7999	9000 ... 9299 9300 ... 9599 9700 ... 9999
40	Power Plant 1 Bark boiler Oil boiler Common services	0001 ... 0299 0300 ... 0599 0600 ... 0999	9000 ... 9299 9300 ... 9599 9600... 9999
42	Power Plant 2	0001 ... 0999	9000...9999
50	RCF 1	0001 ... 0999	9000 ... 9999
52	RCF 2	0001...2999	9000...9999
53	RCF 3	0001...2999	9000...9999
60	Water Treatment Fresh water Effluent	0001 ...0299 0300 ...0599	9000 ... 9299 9300 ... 9599
70	Services and Auxiliary Departments	0001...0999	9000...9999
80	PM2 Process Machine Finishing	0001 ... 2999 3001 ... 3999 7001 ... 7999	9000 ... 9299 9300 ... 9599 9600 ... 9999

NEW System

See Area Sub Process Code Page 57

DSC Loops	001 – 999
Local Gauge	001 – 999

PROJECT STANDARD NO: 21

**Chapter 3 – Instrument Numbering
Appendix XI**

PRINCIPALS AND APPLICATION OF INSTRUMENTATION FUNCTIONAL CODES

CONTENTS

- | | |
|---|--------------------------------|
| 1 | Functional Letter Codes. |
| 2 | Application to Loops. |
| 3 | Application to Device Tagging. |

1 FUNCTIONAL LETTER CODES

The functionality of an instrument loop and identification of items/instruments within a loop is presented by descriptive letter codes.

The letter codes are divided into two groups:

- measured or initial variables.
- function/instrument letters.

The letter codes are presented in Table 1.

Only upper case letters shall be used.

1.1 Group 1 Letters

The first letter shall denote the measured or initiating variable, and shall be in accordance with column 1 of the table, but can be modified, if necessary, by the addition of a letter in accordance with column 2.

Where the variables Q (Quality and Analysis) or E (Electrical) are used then the specific property measured, e.g. O_2 , pH, current, voltage, power etc shall be shown on the P&I diagram by adding an explanatory note outside the symbol circle. Also other instrument identifications can, if required, be provided with explanatory notes outside the symbol circle.

Letter U may be used instead of series of first letters, where a multiplicity of inputs representing dissimilar variables are fed into a single unit.

User's choice letter is Y for unlisted measured or other initiating variables. If used regularly in a particular project, the meaning needs to be defined only once.

The unclassified letter X is intended to cover unlisted measured or other initiating variables that will be used only once or to a limit extent. If used, this letter may have any number of meanings as first letter. The meanings shall be defined outside the symbol circle on P&I diagrams.

1.2 Group 2 Letters

The letters identifying functions are given in column 3 and the letters identifying specific instruments in column 4. They are used as succeeding letters.

If "High" or "Low" require expressing in conjunction with function letters A, O, S or Z, the qualifying letters H or L may be used next to the instrument symbol.

The sign + stands for "On", "Open", "Running" While – stands for "Closed", "Off", "Stopped".

2 APPLICATION TO LOOPS

The following division has been made:

- 1) Loops connected to control system.
- 2) Local loops.

2.1 Loops connected to Control Systems

DCS, MCS, QCS, etc. are understood as control system.

Basically two (2) letters only shall be used to define the functionality. However for expressing:

- ratio, modifier F.
- difference, modifier D.

where required are used thus leading to a three (3) letter definition.

Old example:

20LC-135
└────────── Level Control

20PDC-136
└────────── Pressure Difference Control

New example:

53-20LC-136
└────────── Level Control

53-20PDC-136
└────────── Pressure Difference Control

2.1.1 Control Loop

Letters A, H, I, L, Q, R, S and T are not used as a modifier - given in column 2.

Control loops are called FC, FFC, PC, PDC, QC, LC, etc.

Manual control loops are called HI.

Software control/calculation loops without direct hardware connection are called XC.

2.1.2 Measurement Loops

Letters A, C, H, L, Q, R, S and T are not used as a modifier - given in table 1 column 2.

Measurement loops are called FI, PI, PDI, QI, LI, etc.

2.1.3 Binary Indication Loops

Level, pressure, temperature, limit, position switches, etc.

Letters A, C, H, I, L, Q, R and T are not used as a modifier – given in table 1 column 2.

Binary indication loops are called FS, PS, PDS, QS, LS, etc.

2.1.4 On-off valve Loops

On-off valve loops are always called HS.

No reference is made whether the valve has external limit switches or not.

PROJECT STANDARD NO: 21

Chapter 3 - Instrument Numbering
Appendix XI

2.2 Local Loops

The loop identification contains as many letters as needed.

To be updated by Shotton Mill with new examples

Old example:

20PCV-1135

└────────── Self-Operated Pressure Regulator

New example:

53-20PCV-1135

└────────── Self-Operated Pressure Regulator

For local indicators, recorders, counters, alarm lamps or other hardware devices included in the loop, letters A, H, I, L, Q, R, S and T are used as succeeding letters.

3 APPLICATION TO DEVICE TAGGING

Group 2 \ column 4 letters are used as succeeding letter code to express the kind of instrument in a loop.

Old example:

20LT-0135

└────────── Level Transmitter

New example:

53-20LT0135

└────────── Level Transmitter

For local indicators, recorders, alarm lamps or other hardware devices included in the loop, letters A, H, I, L, Q, R, S and T are used as succeeding letters.

TABLE 1 Identification Letters for Instrumentation

For MCS, additional identification letters for instrumentation will be applied by Valmet

L e t t e r	Group 1		Group 2	
	Measured or initial variable		Function	Instrument
	Meaning as first letter	Meaning as modifier	Meaning as succeeding letter	
	(1)	(2)	(3)	(4)
A	Vibration	-	Alarm	Annunciator
B	-	-	-	-
C	Unclassified variables	-	Controlling	Controller
D	Density, Directional valves	Difference	Quick exhaust	-
E	All electrical variables	-	-	Primary element
F	Flow	Ratio	-	-
G	Gauging, position or length	-	-	-
H	Hand or direct inter-lock operated	-	High	-
I	Monitoring	-	Indicating	Indicator, On-off indication
J	-	Scan	-	-
K	Time or time programme or sequence	-	-	-
L	Level	-	Low	-
M	Moisture or humidity	-	-	Signal converter, proportional valves
N	Consistency	-	-	-
O	-	-	-	Limit switches
P	Pressure or vacuum	-	-	Test point
Q	Quality and analysis, for example pH, conductivity, etc.	Integrate or totalize	Integrating or summing	Integrating or summing device
R	Frequency	-	Recording	Recorder or printer
S	Speed	-	Switching	Switch
T	Temperature	-	Transmitting	Transmitter
U	Multivariable	-	-	Multifunction unit
V	Viscosity	-	-	Valve, damper, etc.
W	Weight or force	-	-	Well, process connection
X	Unclassified variables	-	Unclassified variables	Computing relay or relay
Y	User's choice	-	-	-
Z	-	-	Emergency or safety acting	Emergency or safety device

INSTRUMENTATION SYMBOLS AND PRESENTATION IN P&I DIAGRAM

CONTENTS

- 1 Symbols and Presentation in P&I Diagram.
- 2 Point of Measure.
- 3 Signal Lines.
- 4 Instruments.
- 5 Computer Based Control.
- 6 Distributed Control System.
- 7 Machine Control System.
- 8 Correcting and Actuating Elements.

ANNEXES

- I Instrumentation symbols and application examples.

PROJECT STANDARD NO: 21

Chapter 3 - Instrument Numbering
Appendix XII

1 SYMBOLS AND PRESENTATION IN P&I DIAGRAM

The use of instrumentation symbols in process flow diagrams is presented in Annex 1 in the form of examples.

The symbols will enable users to define and identify in simple form the functions of instruments, the process computer, and distributed control system functions, or any combination of these.

2 POINT OF MEASUREMENT

The symbol for the point of measurement is a small circle of approximately 2 mm diameter located on a line or to a plant equipment outline. The point of measurement is always identified with a tag number as shown in examples. The location of the symbol should be functionally correct and placed in the correct process sequence as close as possible to real location.

3 SIGNAL LINES

The preferred general symbol for an instrument measurement signal line is a continuous thin line.

A thin line with strokes repeated along its length, is used for instrument control signals.

4 INSTRUMENTS

The basic symbol for instruments is a thin line circle of approximately 14 mm diameter. The letter code with its identification number is placed inside the circle. If the letter code or identification number cannot be accommodated within the circle, the circle may be broken.

The panel mounted instrument is indicated as instrument symbol added with a horizontal thin line across it.

The symbol for locally mounted instrument has no horizontal line.

If required, the instrument mounted on a subpanel can be indicated with two horizontal lines.

The transmitters are normally not indicated in process flow diagrams. If, however, the same receiver instrument measures a number of different physical properties, their transmitters can be indicated.

The area and sub area codes will always appear left side of the circle.

PROJECT STANDARD NO: 21

Chapter 3 - Instrument Numbering
Appendix XII

5 COMPUTER BASED CONTROL

If the instrument loop is connected to Process Computer (PC), the basic symbol will be equilateral hexagon approximately same size as the basic symbol for instruments.

The horizontal line is used when there is indication or control in PC and two horizontal lines when there is connection to local panel.

The area and sub area codes will always appear left side of the circle.

6 DISTRIBUTED CONTROL SYSTEM

If the instrument loop is connected to Distributed Control System (DCS), the basic symbol will be a square with circle inside same size as the basic symbol for instruments.

The horizontal line is used when there is indication or control in DCS and two horizontal lines when there is connection to local panel.

The area and sub area codes will always appear left side of the circle.

7 MACHINE CONTROL SYSTEM

If the instrument loop is connected to Machine Control System (MCS), the basic symbol will be a equilateral diamond with circle inside same size as the basic symbol for instruments.

The horizontal line is used when there is indication or control in PLS and two horizontal lines when there is connection to local panel.

The area and sub area codes will always appear left side of the circle.

8 CORRECTING AND ACTUATING ELEMENTS

The symbol for a correcting element of unspecified type is an equilateral triangle with sides of approximately 5 mm length.

If the type of the correcting element is known, the corresponding symbol can be used, e.g. that for a valve.

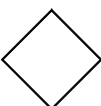
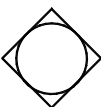
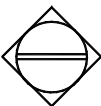
The basic symbol for an automatic actuating element is a thin line circle of approximately 5 mm diameter, with a thin line to connect it to the correcting element symbol.

[illegible]

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE
CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
2/14

DESCRIPTION	SYMBOL	NOTES
5.1 BASIC SYMBOL FOR PROCESS COMPUTER (PC)		PROCESS COMPUTER ONLY
5.2 INDICATION OR CONTROL IN PC		PROCESS COMPUTER ONLY
5.3 CONNECTION TO LOCAL PANEL		PROCESS COMPUTER ONLY
5.4 BASIC SYMBOL FOR DISTRIBUTED CONTROL SYSTEM (DCS)		DISTRIBUTED CONTROL SYSTEM ONLY
5.5 INDICATION OR CONTROL IN DCS		DISTRIBUTED CONTROL SYSTEM ONLY
5.5 CONNECTION TO LOCAL PANEL		DISTRIBUTED CONTROL SYSTEM ONLY
5.7 LOGIC AND SEQUENTIAL FUNCTION		
5.8 BASIC SYMBOLE FOR MACHINE CONTROL SYSTEM (MCS)		MCS ONLY
5.9 INDICATION OR CONTROL IN MCS		MCS ONLY
5.10 CONNECTION TO LOCAL PANEL		MCS ONLY
5.11 CORRECTING ELEMENT		
5.12 AUTOMATIC ACTUATING ELEMENT		

JP No	16A1605
Dwg No	
Sheet	2
Loop No	
Superseded	
Department	
Type	I- STANDARDS
Application	INSTRUMENTATION SYMBOLS AND APPLICATION EXAMPLES
Archive Code	
Binder Code	
Drawn	27.10.99 pcs
Designed	27.10.99 JKS
Checked	
Approved	
 SHOTTON PAPER UPM	
	
90	

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

[illegible]

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
4/14

DESCRIPTION	SYMBOL	NOTES
5.20 CORRECTING UNIT		FOR EXAMPLE: SPEED ADJUSTING
5.21 MANUAL VALVE		NORMALLY OPEN
5.22 MANUAL VALVE		NORMALLY CLOSED
5.23 SAFETY RELEASE VALVE		IN GENERAL
5.24 CONTROL VALVE		HAND OPERATED
5.25 CONTROL AND ON-OFF VALVE		IN GENERAL
5.26 CONTROL AND ON-OFF VALVE		MOTOR OPERATED
5.27 CHECK VALVE		IN GENERAL
5.28 PRESSURE REDUCING VALVE		IN GENERAL: FOR EXAMPLE: STEAM PRESSURE REDUCING VALVE
5.29 THREE-WAY VALVE		
5.30 FOUR-WAY VALVE		
5.31 SAFETY ACTION OF CONTROL OR ON-OFF VALVE		FC=FAILURE CLOSE
5.32 SAFETY ACTION OF CONTROL OR ON-OFF VALVE		FO=FAILURE OPEN

JP No 16A1605

Dwg No

Sheet 4

Loop No

Superseded

Department

I-STANDARDS

INSTRUMENTATION SYMBOLS AND
APPLICATION EXAMPLES

Superseded

Archive Code

SHOTTON
PAPER
UPM

Binder Code

Drawn 27.10.99 pcs

Designed 27.10.99 JKS

Checked

Approved

AAKKO PÖYRY

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
5/14

6 INSTRUMENTATION SYMBOLS AND IDENTIFICATION IN FLOW SHEETS,EXAMPLES 6.1 CONTROL LOOPS

DESCRIPTION	SYMBOL	NOTES
6.1.1 FLOW CONTROL ON DCS 20FC-0304		
6.1.2 FLOW CONTROL ON DCS 20FC-0305		
6.1.3 MANUAL VALVE CONTROL ON DCS 20HI-0401		
6.1.4 MANUAL VALVE CONTROL ON DCS VALVE WITH LIMIT SWITCHES INDICATION ON DCS 20HI-0404		
6.1.5 FLOW RATIO		FLOW CONTROLLER 0602 HOLDS A CONSTANT RATIO BETWEEN BOTH FLOW RATES

JP No 16A1605

Dwg No

Sheet 5

Loop No

Superseded

Department

Type

I-STANDARDS
INSTRUMENTATION SYMBOLS AND
APPLICATION EXAMPLES

Supersedes

Archive Code

SHOTTON
PAPER
UPM

Binder Code

Drawn 27.10.99 pcs

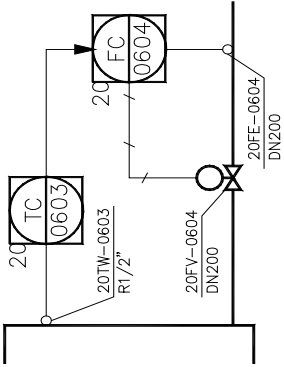
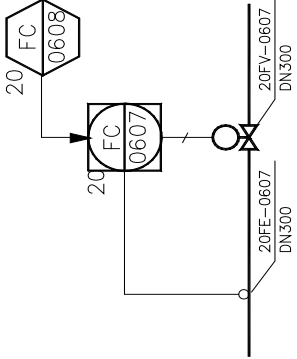
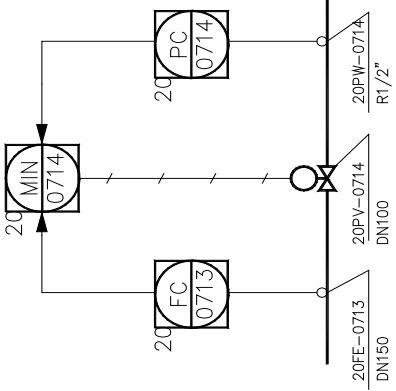
Designed 27.10.99 JKS

Checked

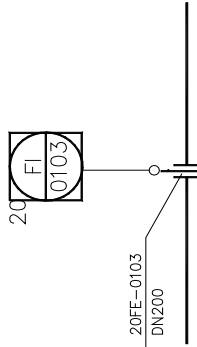
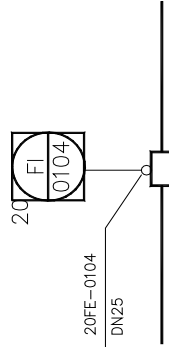
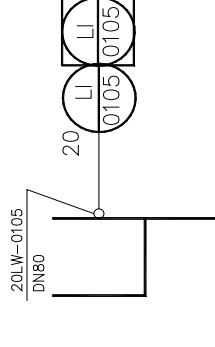
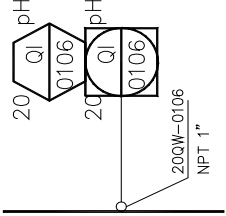
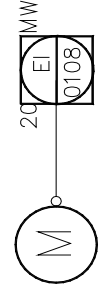
Approved

JAAKKO PÖYRY

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING		APPENDIX XII/ANNEX I 6/14	
DESCRIPTION	SYMBOL	NOTES	JP No 16A1605
6.1.6 CASCADE CONTROL ON DCS		TEMPERATURE CONTROLLER 0603 PROVIDE THE COMMAND SIGNAL TO THE FLOW CONTROLLER 0604	
6.1.7 THE SET POINT OF A DCS CONTROLLER IS REMOTE ADJUSTED FROM PC			
6.1.8 MULTIVARIABLE CONTROL UNIT ON DCS		SELECTS FOR THE CORRECTING UNIT THE LOWER ONE OF TWO SIGNALS	

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING		APPENDIX XII/ANNEX I 7/14	
6 INSTRUMENTATION SYMBOLS AND IDENTIFICATION IN FLOW SHEETS,EXAMPLES 6.2 MEASUREMENT LOOPS			
DESCRIPTION	SYMBOL	NOTES	
6.2.1 FLOW INDICATION ON 20FI–0103		20FI–0103:FLOW SENSOR ORIFICE PLATE	
6.2.2 FLOW INDICATION ON DCS 20FI–0104		20FI–0104:FLOW SENSOR MAGNETIC FLOW METER	
6.2.3 LEVEL, INDICATION ON CONTROL PANEL AND DCS 20LI–0105		20LI–0105: PROCESS CONNECTION FLANGE DN80	
6.2.4 pH INDICATION ON DCS AND PC 20QI–0106		20QI–0106: THREADED NPT 1" PROCESS CONNECTION	
6.2.5 ELECTRIC POWER INDICATION ON DCS			
UP No 16A1605		Department	
Dwg No		Type	
Sheet 7		Application	
Superseded		I-STANDARDS INSTRUMENTATION SYMBOLS AND APPLICATION EXAMPLES	
		Supersedes	
		M	
		SHOTTON UPM PAPER	
		Archive Code	
		Binder Code	
		Drawn 27.10.99 pcs	
		Designed 27.10.99 JKS	
		Checked	
		Approved	
		96	
		AANKO PORYAY	

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE
CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XI/ANNEX II
8/14

DESCRIPTION	SYMBOL	NOTES
6.2.6 2 PRESSURE MEASUREMENTS, INDICATION ON DCS CALCULATED DIFFERENTIAL PRESSURE		
6.2.7 PRESSURE MEASUREMENTS IN DCS 20PI-0351		
6.2.8 FLOW AND CONSISTENCY MEASUREMENTS, INDICATION ON DCS, CALCULATED PRODUCTION		
6.2.9 LEVEL MEASUREMENT PRESSURIZED TANK, INDICATION ON DCS		
6.2.10 FLOW MEASUREMENT PROVIDED WITH PRESSURE AND TEMPERATURE COMPENSATION ON DCS		

QP No 16A1605

Dwg No

Sheet 8

Loop No

Superseded

Department

Type

Application

I-STANDARDS
INSTRUMENTATION SYMBOLS AND
APPLICATION EXAMPLES

Supersedes

Archive Code

SHOTTON
PAPER
UPM

Binder Code

Drawn 27.10.99 pcs

Designed 27.10.99 JKS

Checked

Approved

AAKKO PÖYRY

CHAPTER 3 INSTRUMENTATION No's APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
9/14

6 INSTRUMENTATION SYMBOLS AND IDENTIFICATION IN FLOW SHEETS,EXAMPLES 6.3 BINARY INDICATION LOOPS

DESCRIPTION	SYMBOL	NOTES
6.3.1 MANUAL VALVE WITH LIMIT SWITCHES, INDICATION ON DCS 20GO-0204		
6.3.2 PRESSURE SWITCH, ALARM ON DCS 20PS-0205		
6.3.3 HIGH AND LOW LEVEL SWITCHES, ALARMS ON DCS		
6.3.4 LOCAL DIFFERENTIAL PRESSURE GAUGE WITH DIFF.PRESSURE SWITCH, ALARMS ON DCS		

UP No 16A1605
Dwg No
Sheet 9
Loop No

Department
Type
Application
I-STANDARDS
INSTRUMENTATION SYMBOLS AND
APPLICATION EXAMPLES

Archive Code

SHOTTON
UPM PAPER

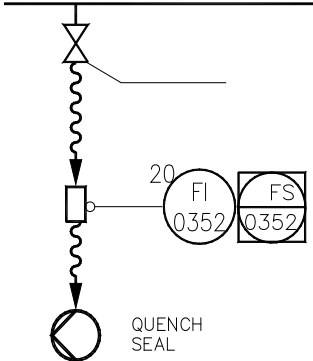
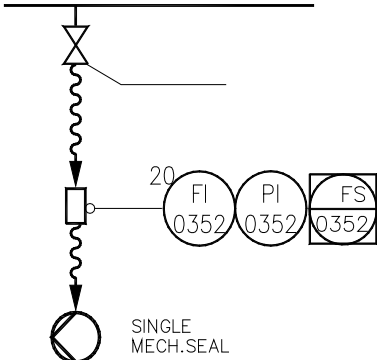
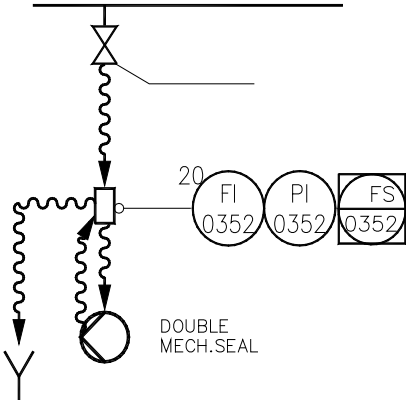
Binder Code
Drawn 27.10.99 pcs
Designed 27.10.99 JKS
Checked
Approved

JAAKKO PÖYRY

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE
CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
10/14

			UP No 16A1605
			Dwg No
			Sheet 10
			Loop No
			Superseded
DESCRIPTION	SYMBOL	NOTES	
6.3.5 SEALING WATER ROTAMETER WITH FLOW SWITCH ALARM TO DCS 20FS-0352			I-STANDARDS INSTRUMENTATION SYMBOLS AND APPLICATION EXAMPLES
			Department Type Application
6.3.6 SEALING WATER ROTAMETER WITH FLOW SWITCH ALARM TO DCS 20FS-0352			Archive Code SHOTTON PAPER UPM
			Binder Code
			Drawn 27.10.99 pcs
			Designed 27.10.99 JKS
			Checked
			Approved
			JAAKKO PÖYRY
			99

CHAPTER 3 INSTRUMENTATION No's APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
11/14

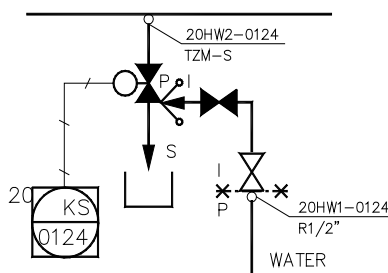
6 INSTRUMENTATION SYMBOLS AND IDENTIFICATION IN FLOW SHEETS,EXAMPLES 6.4 ON-OFF VALVE LOOPS

DESCRIPTION

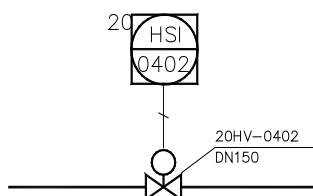
SYMBOL

NOTES

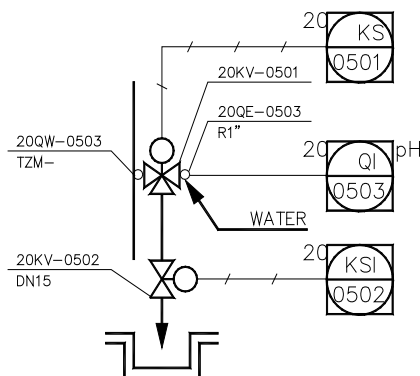
6.4.1
DCS OPERATED
SAMPLING VALVE



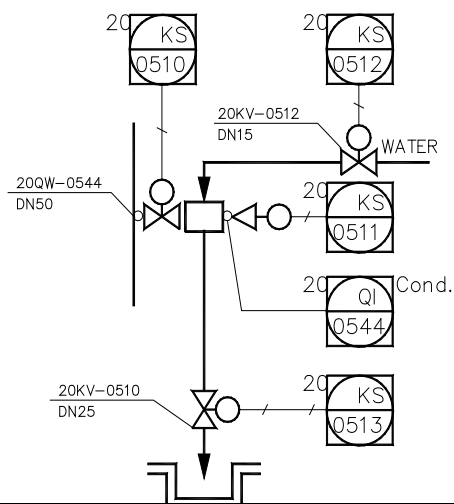
6.4.2
ON-OFF VALVE WITH
LIMIT SWITCHES
MANUALLY CONTROLLED
FROM DCS



6.4.3
pH MEASUREMENT
WITH AUTOMATIC
FLUSHING,
DCS OPERATED



6.4.4
CONDUCTIVITY MEASUREMENT
WITH AUTOMATIC
FLUSHING,
DCS OPERATED



JP No 16A1605

Dwg No

Sheet 11

Loop No

Superseded

Department

Type

Application

Instrumentation Symbols and Application Examples

Archive Code

SHOTTON

PAPER

UPM

Binder Code

Drawn 27.10.99 pcs

Designed 27.10.99 JKS

Checked

Approved

JAAKKO PÖYRY

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING		APPENDIX XII/ANNEX I 12/14
DESCRIPTION	SYMBOL	NOTES
6.4.5 TIME-CYCLE OPERATION IN DCS		
Department		Archive Code
Type		Binder Code
I-STANDARDS		UPM PAPER SHOTTON
Application		Checked
INSTRUMENTATION SYMBOLS AND APPLICATION EXAMPLES		Designed
Supersedes		27.10.99 pcs
Superseded		Drawn
Loop No		27.10.99
Sheet		27.10.99 JKS
Dwg No		Approved
JP No 16A1605		101

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
13/14

6 INSTRUMENTATION SYMBOLS AND IDENTIFICATION IN FLOW SHEETS,EXAMPLES 6.5 LOCAL LOOPS

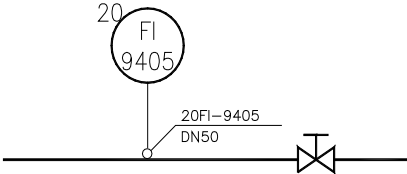
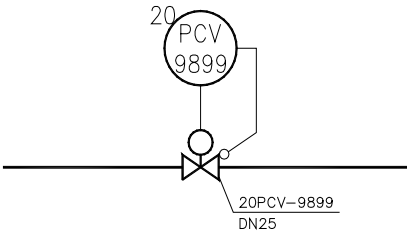
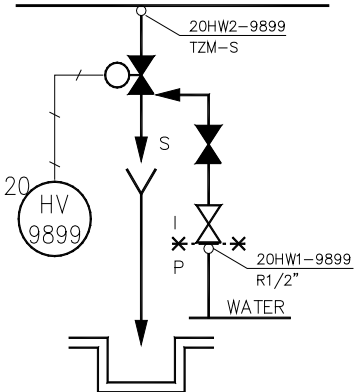
DESCRIPTION	SYMBOL	NOTES
6.5.1 PRESSURE, LOCAL INDICATION 20PI-9101		
6.5.2 DIFFERENTIAL PRESSURE, INDICATION ON CONTROL PANEL 20PDI-9102		
6.5.3 TEMPERATURE, POINT OF MEASUREMENT IN THE PLANT EQUIPEMENT 20T-9109		
6.5.4 LOCALLY OPERATED PNEUMATIC SAMPLING VALVE 20HS-9123		
6.5.5 CLOSED CIRCUIT TV CAMERA		

JP No	16A1605
Dwg No	
Sheet	13
APP No	APP_2_13.dwg 03.11.1999 08: /PCS
Superseded	
Department	I-STANDARDS
Type	INSTRUMENTATION SYMBOLS AND APPLICATION EXAMPLES
Archive Code	SHOTTON UPM PAPER
Blinder Code	
Drawn	27.10.99 pcs
Designed	27.10.99 JKS
Checked	
Approved	
AAKKO PÜYRY	102

CHAPTER 3 INSTRUMENTATION No'S APPENDIX XII

PLANT NUMBERING PROCEDURE
CHAPTER 3 – INSTRUMENT NUMBERING

APPENDIX XII/ANNEX I
14/14

DESCRIPTION			SYMBOL	NOTES
6.5.6 LOCAL CONTROL WITH HAND OPERATED VALVE AND LOCAL INDICATOR				
6.5.7 SELF OPERATED PRESSURE REGULATOR 20PCV-9899				
6.5.8 MANUAL SAMPLING VALVE 20HV-9899				

JP No	16A1605
Dwg No	
Sheet	14
Loop No	
Superseded	

Department	I-STANDARDS
Type	INSTRUMENTATION SYMBOLS AND APPLICATION EXAMPLES
Application	
Supersedes	

Archive Code	SHOTTON UPM PAPER
Binder Code	

Drawn	27.10.99	pcs
Designed	27.10.99	JKS
Checked		
Approved		

JAAKKO PÖYRY	103
--------------	-----

PROJECT STANDARD NO: 21

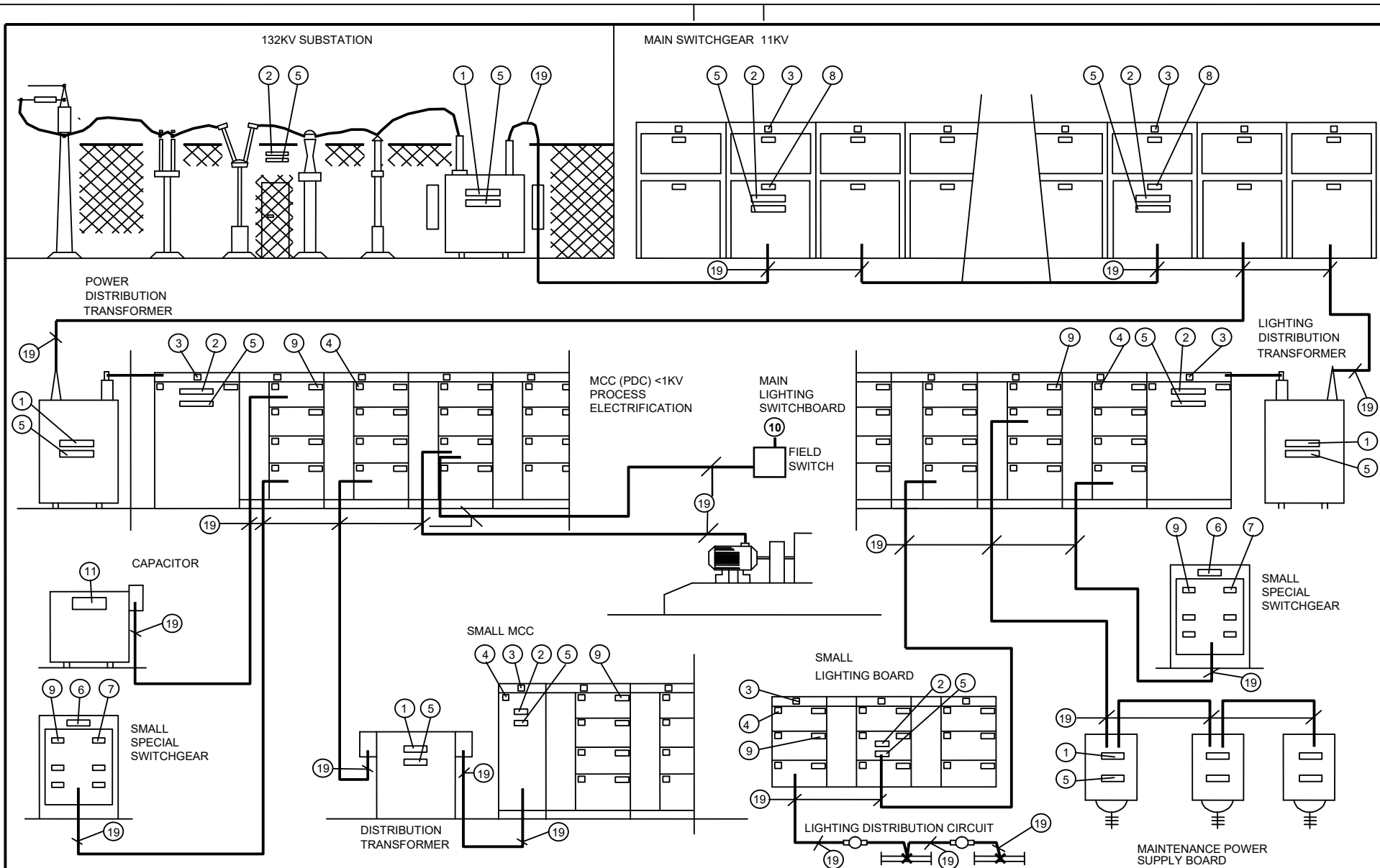
**Chapter 4 – MARKING AND NAMEPLATES FOR
INSTRUMENTATION AND ELECTRIFICATION**

PLANT NUMBERING PROCEDURE

Chapter 4 – MARKING AND NAMEPLATES FOR INSTRUMENTATION AND ELECTRIFICATION

For new Projects post 23/03/2022

[illegible]



CHECKED

DRAWN I.H. 18 -1-96

DESIGN



SHTTON
PAPER

DEPARTMENT

TYPE

NAME

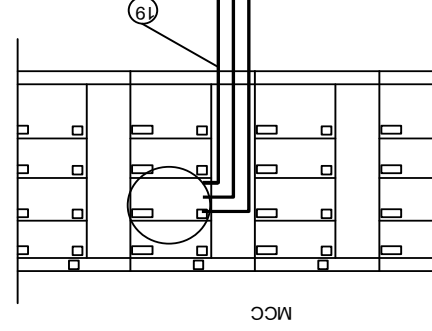
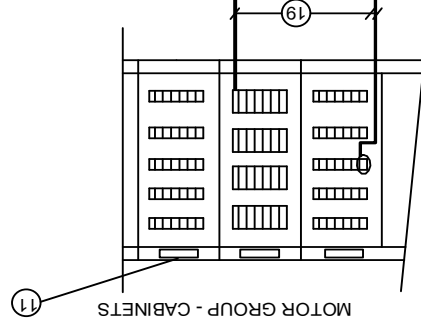
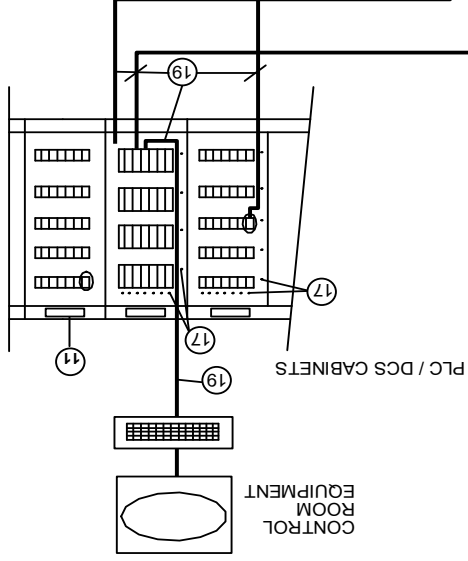
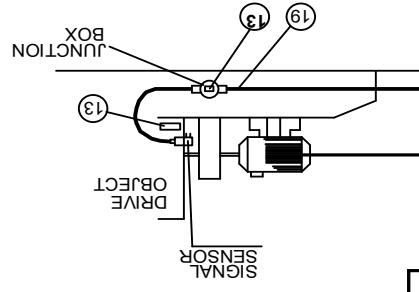
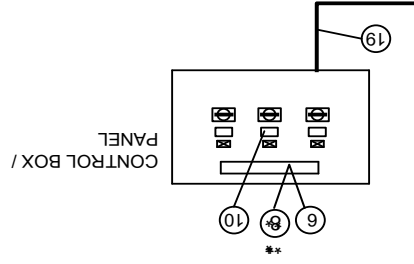
THE POWER DISTRIBUTION EQUIPMENT

CHAPTER 4

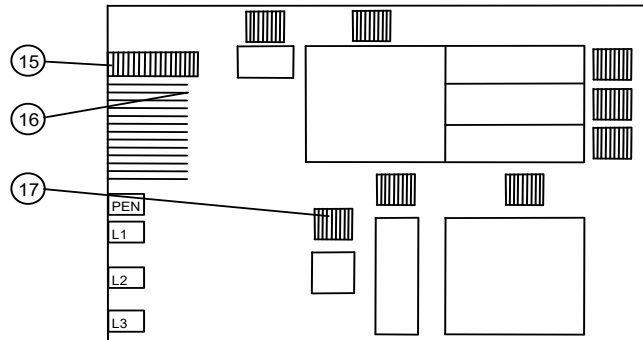
2

PROJECT STANDARD No. 31

* DEPENDING ON THE SIZE



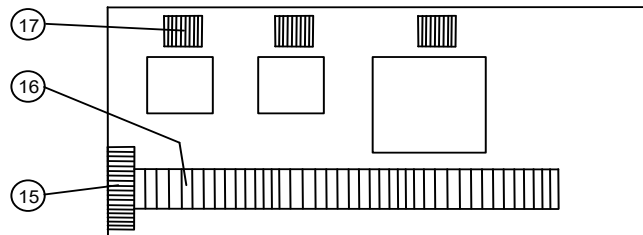
SWITCHGEAR AND MCCS



INTERNAL WIRING OF THE STARTER
SHALL BE MARKED

INTERNAL CONNECTIONS TO OTHER MODULES
WILL BE MARKED AS SHOWN ON THE WIRING DIAGRAM
BY USING CONDUCTOR MARKERS, TYPE 45

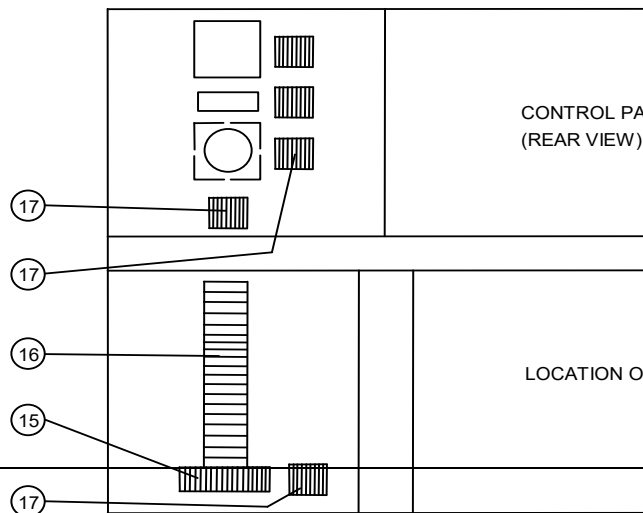
RELAY RACKS



INTERNAL WIRING OF THE COMPONENT RELAY RACK
MODULE SHALL BE MARKED

INTERNAL CONNECTIONS TO OTHER MODULES
WILL BE MARKED AS SHOWN ON THE WIRING DIAGRAM
BY USING CONDUCTOR MARKERS, TYPE 45

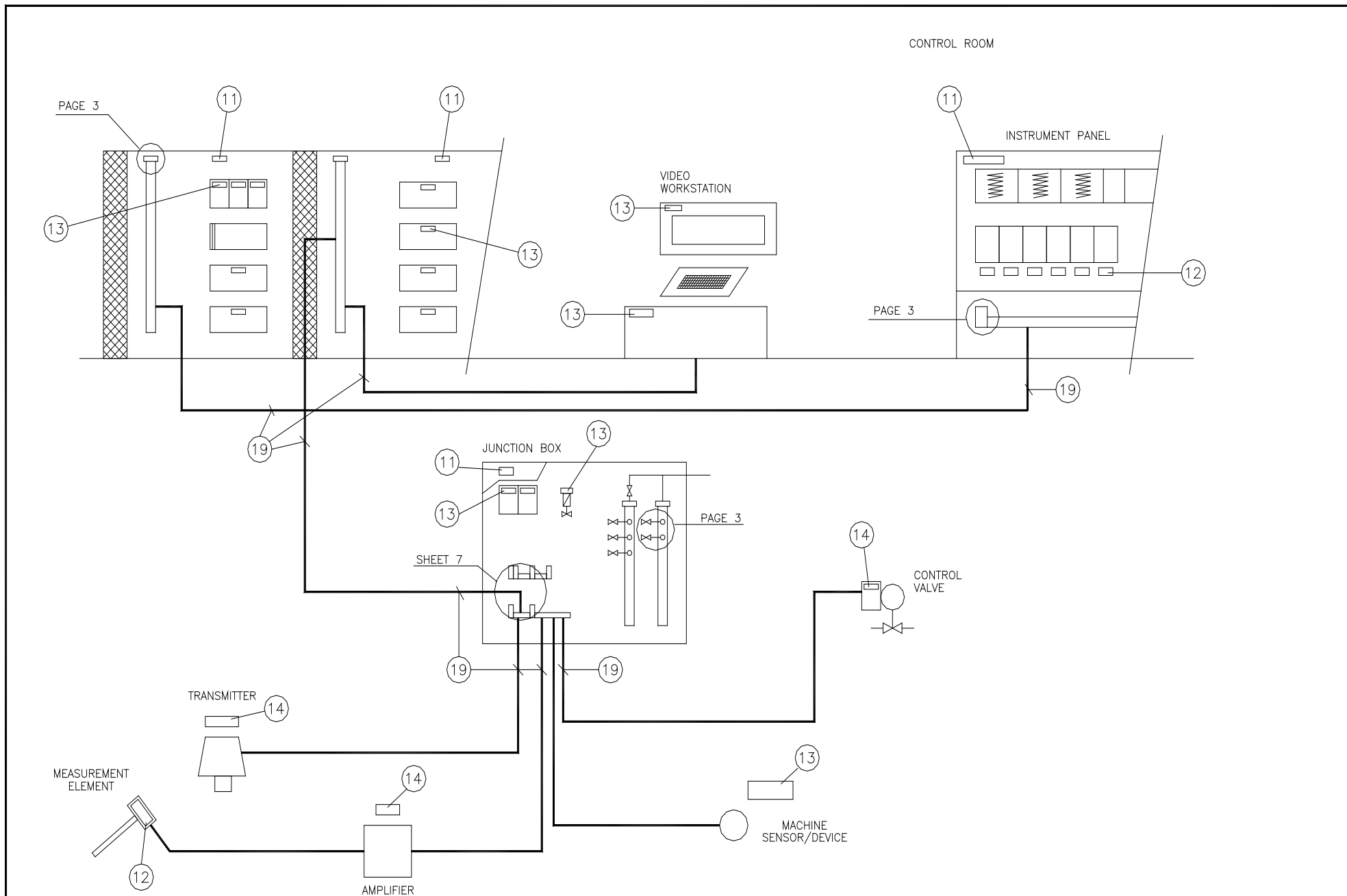
CONTROL PANEL



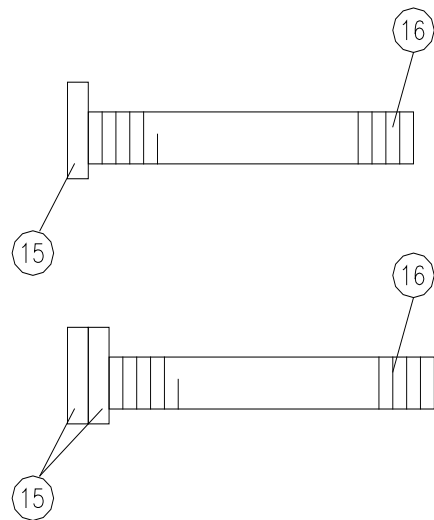
CONTROL PANEL
(REAR VIEW)

LOCATION OF TERMINALS

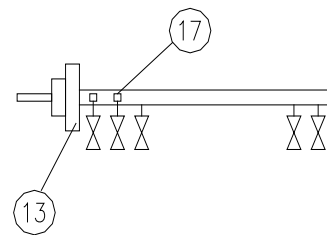
INTERNAL WIRING OF THE CONTROL STATION
WILL BE MARKED AS SHOWN ON THE WIRING DIAGRAM
BY USING CONDUCTOR MARKERS, TYPE 45



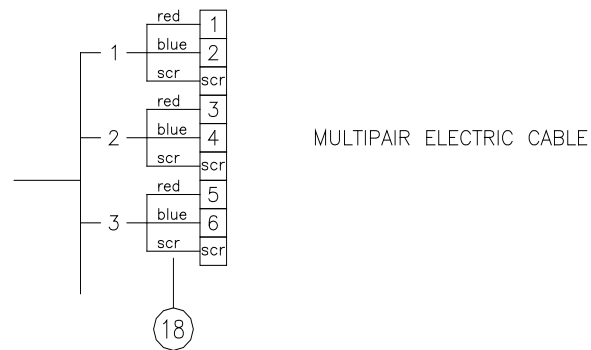
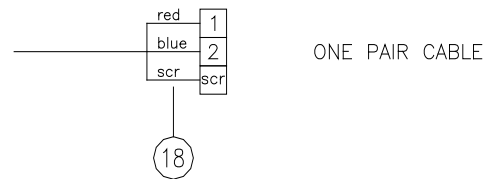
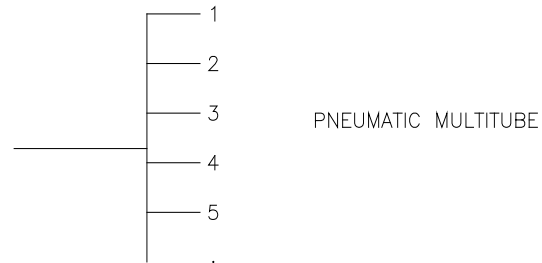
TERMINALS



AIR DISTRIBUTION

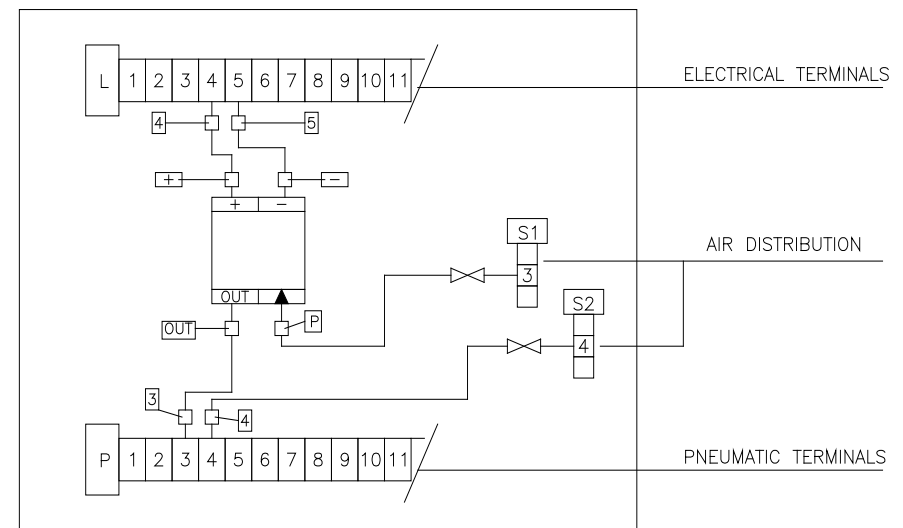


MARKING OF CABLE CONDUCTORS



INTERNAL CONNECTIONS INSIDE
INSTRUMENT PANELS, RACKS AND
JUNCTION BOXES SHALL BE MARKED
AS SHOWN ON THE WIRING DIAGRAM
BY USING CONDUCTOR MARKERS.

EXAMPLE:
MARKING IN JUNCTION BOX



15



TERMINAL STRIPE NUMBER OR
ELECTRIC LOOP NUMBER

TERMINAL MARKER TYPE
BY BOX MANUFACTURER

16



TERMINAL NUMBER

TERMINAL MARKER TYPE
BY BOX MANUFACTURER

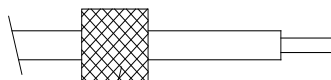
17



VALVE NUMBER/LOCATION NUMBER
ITEM NUMBER OF COMPONENT

BY JUNCTION/SV BOX MANUFACTURER

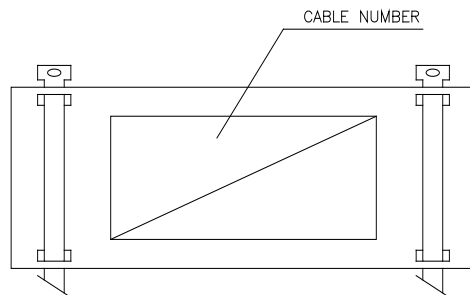
18



NUMBER OF THE CABLE
CONDUCTOR OR WIRE

CABLE MARKERS BY INSTALLATION
CONTRACTOR

19



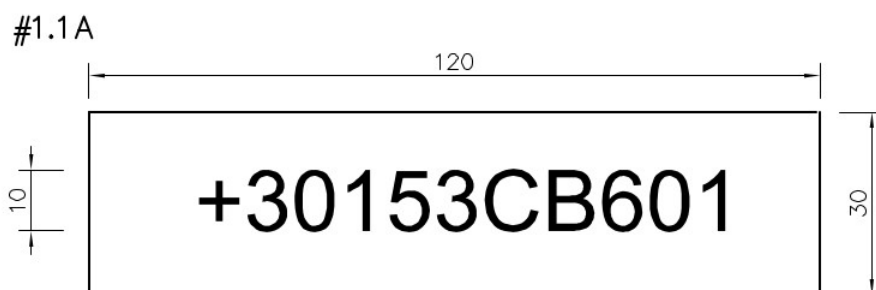
CABLE NUMBER

CABLE MARKERS BY INSTALLATION
CONTRACTOR

NAME PLATE	COLOUR	COLOUR OF THE TEXT	FASTENING	NOTE
15	WHITE	BLACK	TERMINAL BLOCK MARKER	
16	WHITE	BLACK	ON THE TERMINAL	
17	WHITE	BLACK	SELF ADHESIVE	PRINTED TYPE
18	YELLOW	BLACK	ROUND THE CONDUCTOR OR WIRE	
19	YELLOW	BLACK	WIRE OR TIE WRAP	WEATHER PROOF

DRAFT: 2.3 Automation Name Plates Standard

PAP0000594#1.1A CONTROL DESKS/BOXES, TERMINAL BOXES, VALVE PANELS, HYDRAULIC UNITS



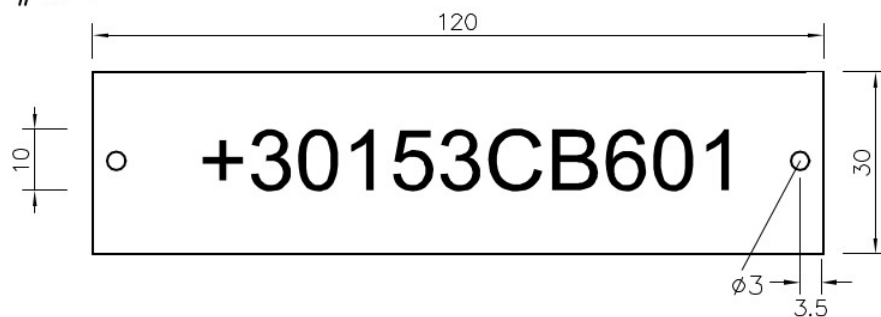
- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape

Usage:

- Dry Areas and Control Boxes with Protective Covers in Wet Areas
- Max 11 characters

PAP0000594#1.1B CONTROL DESKS/BOXES, TERMINAL BOXES, VALVE PANELS, HYDRAULIC UNITS

#1.1B



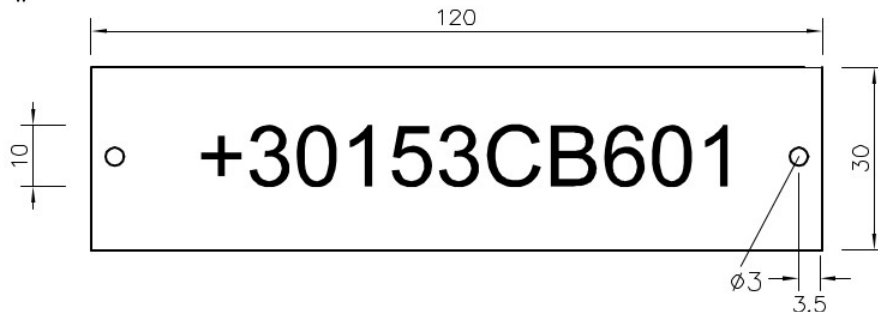
- White plastic, engraving on the back side, black text
- Fastening with thin double-sided tape

Usage:

- Dry Areas and Control Boxes with Protective Covers in Wet Areas
- Max 11 characters

PAP0000594#1.2 CONTROL DESKS/BOXES, TERMINAL BOXES, VALVE PANELS, HYDRAULIC UNITS

#1.2



- Acid-proof steel, laser printing, black text
- Acid-proof screw fastening

Usage:

- Hot or Wet Areas
- Max 11 characters

PAP0000594#2.1A ADDITIONAL PLATES: VALVE PANELS AND HYDRAULIC UNITS

#2.1A



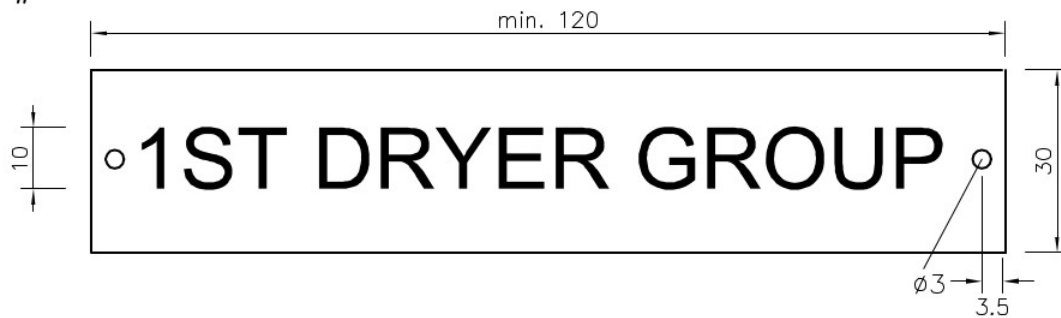
- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape
- Plate min width 120mm, otherwise according to characters

Usage:

- Dry Areas. Install above plate #1.1A

PAP0000594#2.1B ADDITIONAL PLATES: VALVE PANELS AND HYDRAULIC UNITS

#2.1B



- White plastic, engraving on the back side, black text
- Fastening with thin double-sided tape
- Plate min width 120mm, otherwise according to characters

Usage:

- Dry Areas. Install above plate #1.1B

PAP0000594#2.2 ADDITIONAL PLATES: VALVE PANELS AND HYDRAULIC UNITS

#2.2



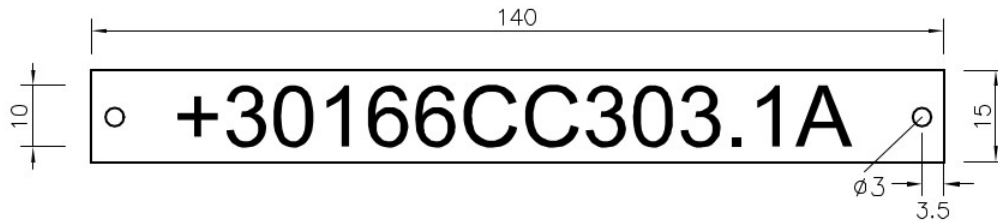
- Acid-proof steel, laser printing, black text
- Acid-proof screw fastening
- Plate min width 120mm, otherwise according to characters

Usage:

- Hot or Wet Areas. Install above plate #1.2

PAP0000594#3.1 FOR DNA CABINETS IP 20 FRAME ONLY

#3.1



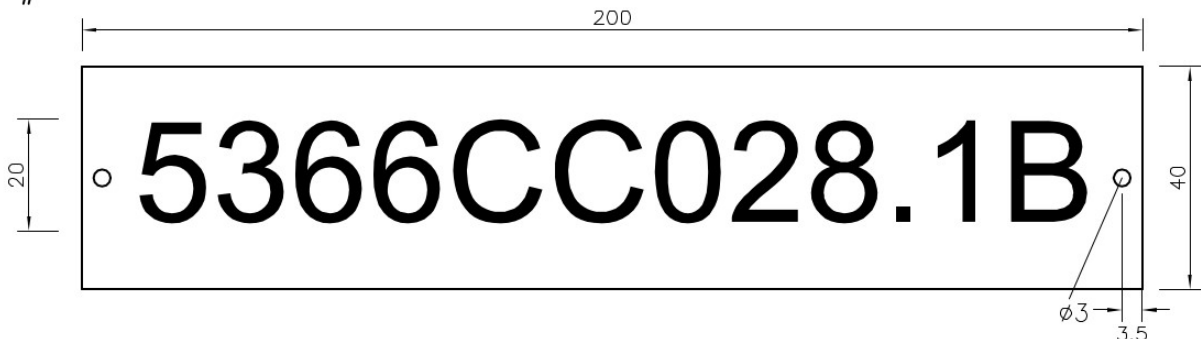
- White plastic, engraving on the back side, black text
- Screw fastening on the top doorframe
- Fastening with thin double-sided tape

Usage:

- Used for old model DNA-cabinets only
- Max 14 characters

PAP0000594#3.2 FOR ALL OTHER PLC- AND DNA-CABINETS

#3.2



- White plastic, engraving on the back side, black text
- Fastening with thin double-sided tape

Usage:

- Used for Rittal cabinets etc.
- Max 14 characters

PAP0000594#4.1 CONTROL EQUIPMENT IN DESKS AND CONTROL BOXES

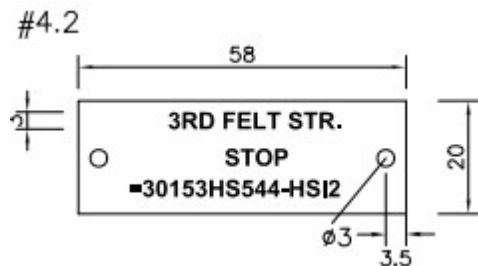


- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape

Usage:

- Dry and Wet Areas
- Max 23 characters each row

PAP0000594#4.2 CONTROL EQUIPMENT IN DESKS AND CONTROL BOXES

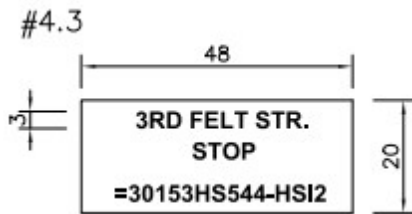


- Acid-proof steel, laser printing, black text
- Acid-proof screw or drive rivet fastening

Usage:

- Hot Areas
- Max 23 characters each row

PAP0000594#4.3 CONTROL EQUIPMENT IN DESKS AND CONTROL BOXES



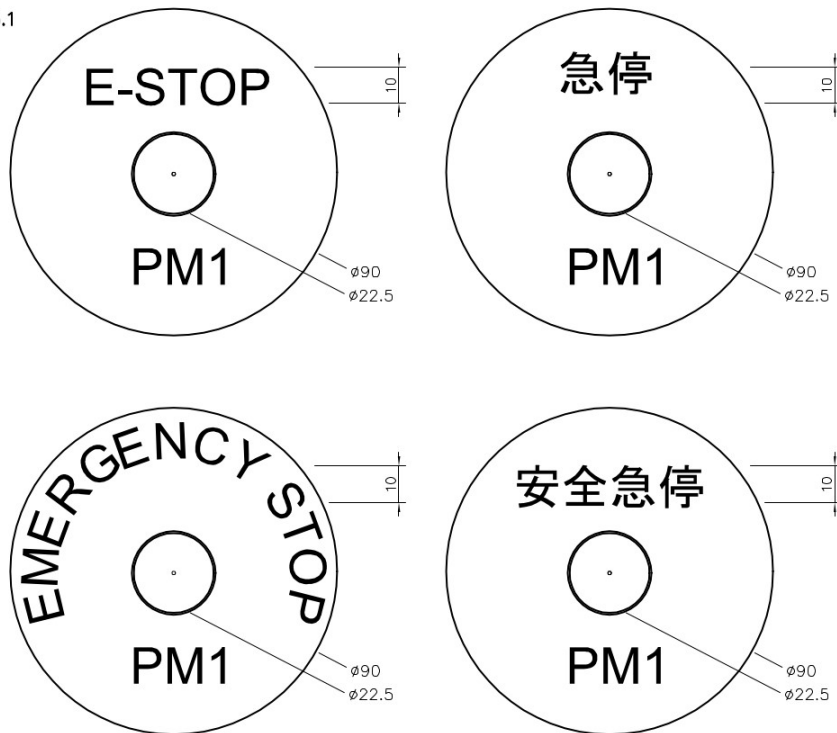
- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape

Usage:

- Dry and Wet Areas
- Max 18 characters each row

PAP0000594#5.1 EMERGENCY STOP PUSHBUTTON

#5.1

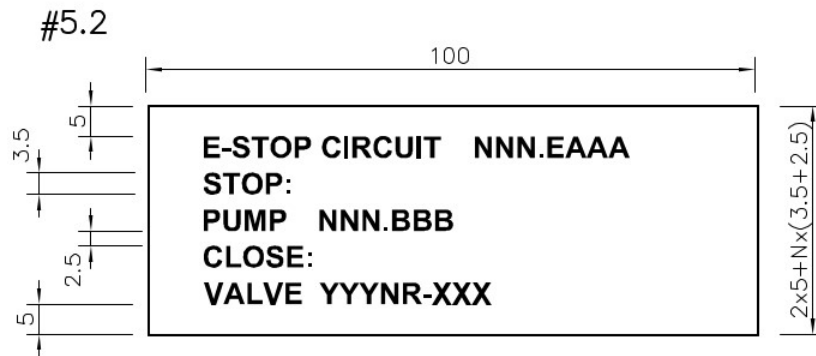


- Yellow plastic, engraving on the top side, black text
- Text in English: E-STOP / EMERGENCY STOP
- Text in Chinese: 急停 / 安全急停

Usage:

- Max 14 characters top row
- Max 6 characters bottom row

PAP0000594#5.2 ADDITIONAL PLATES



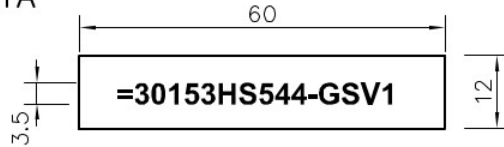
- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape
- Text in English: E-STOP CIRCUIT
- Text in English: STOP:
- Text in English: CLOSE:

Usage:

- An additional plate is required only in special cases where the process has been divided into emergency stop zones and the emergency stop area cannot be indicated in any other way
- Max 30 characters each row

PAP0000594#6.1A ELECTRICAL, HYDRAULIC AND PNEUMATIC EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS AND ON MACHINE FRAMES

#6.1A



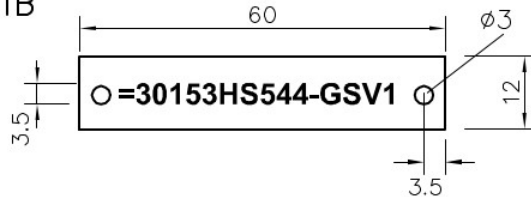
- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape

Usage:

- Dry Areas
- Max 22 characters

PAP0000594#6.1B ELECTRICAL, HYDRAULIC AND PNEUMATIC EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS AND ON MACHINE FRAMES

#6.1B

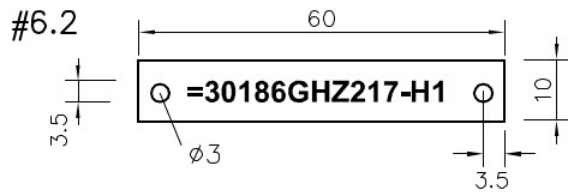


- White plastic, engraving on the back side, black text
- Fastening with thin double-sided tape

Usage:

- Dry Areas
- Max 19 characters

PAP0000594#6.2 ELECTRICAL, HYDRAULIC AND PNEUMATIC EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS AND ON MACHINE FRAMES

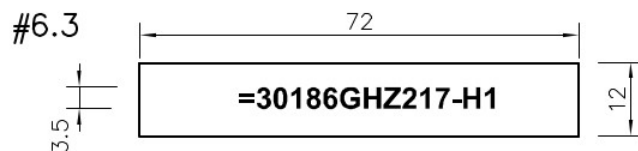


- Acid-proof steel, laser printing, black text
- Acid-proof screw or drive rivet fastening

Usage:

- Hot or Wet Areas
- Max 19 characters

PAP0000594#6.3 ELECTRICAL, HYDRAULIC AND PNEUMATIC EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS AND ON MACHINE FRAMES

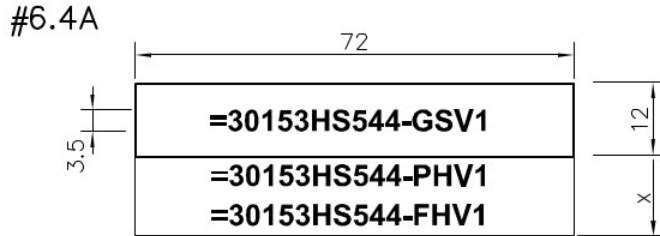


- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape

Usage:

- Valve Cabinets and Boxes
- Max 25 characters

PAP0000594#6.4A ELECTRICAL, HYDRAULIC AND PNEUMATIC EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS AND ON MACHINE FRAMES

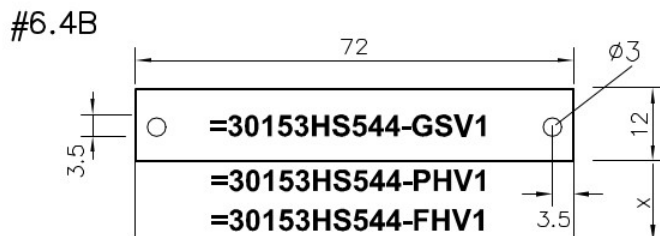


- White plastic, engraving on the back side, black text
- The number of device position rows equals to the number of valves
- Height according to the number of rows
- Fastening with acrylic foam tape

Usage:

- Valve Cabinets and Boxes, Sandwich Valve Blocks
- Max 25 characters each row

PAP0000594#6.4B ELECTRICAL, HYDRAULIC AND PNEUMATIC EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS AND ON MACHINE FRAMES

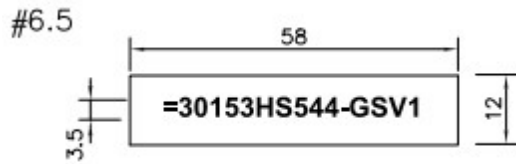


- White plastic, engraving on the back side, black text
- The number of device position rows equals to the number of valves
- Height according to the number of rows
- Fastening with thin double-sided tape and screw or drive rivet fastening

Usage:

- Valve Cabinets and Boxes, Sandwich Valve Blocks
- Max 25 characters each row

**PAP0000594#6.5 ELECTRICAL, HYDRAULIC AND PNEUMATIC
EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC
UNITS AND ON MACHINE FRAMES**

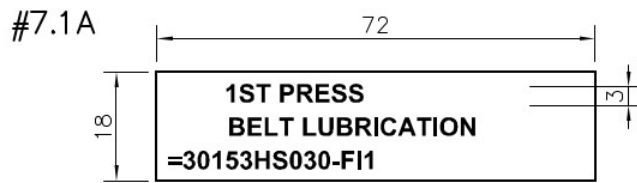


- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape

Usage:

- Dry Areas
- Max 20 characters

PAP0000594#7.1A METERS AND INDICATOR EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS

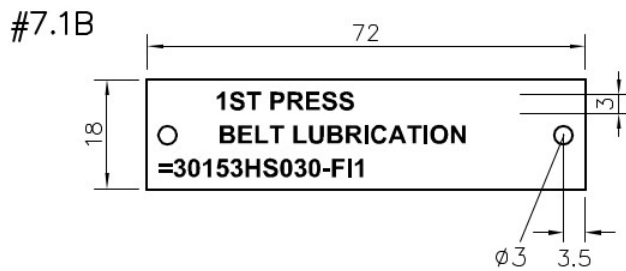


- White plastic, engraving on the back side, black text
- Fastening with acrylic foam tape

Usage:

- Dry Areas
- Max 30 characters

PAP0000594#7.1B METERS AND INDICATOR EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS



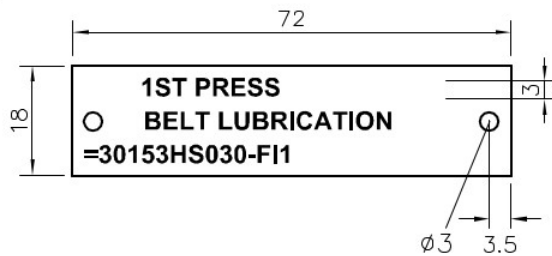
- White plastic, engraving on the back side, black text
- Fastening with thin double-sided tape

Usage:

- Dry Areas
- Max 30 characters

PAP0000594#7.2 METERS AND INDICATOR EQUIPMENT INSIDE BOXES AND VALVE PANELS AND IN HYDRAULIC UNITS

#7.2



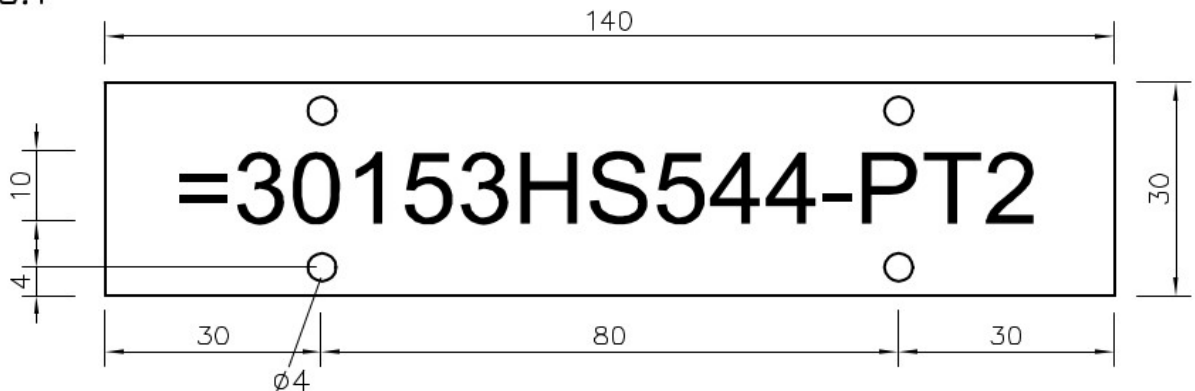
- Acid-proof steel, laser printing, black text
- Acid-proof screw fastening.

Usage:

- Hot or Wet Areas
- Max 30 characters

PAP0000594#8.1 FIELD EQUIPMENT, ON PROCESS PIPING AND MACHINE FRAME

#8.1



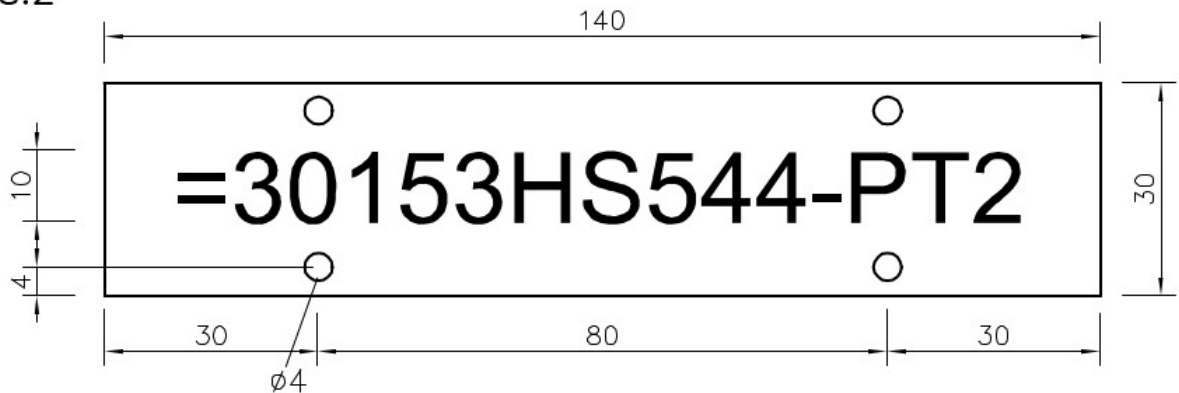
- White plastic, engraving on the back side, black text
- Acid-proof steel wire or cable tie (EN 1.4404 / AISI 316 L)

Usage:

- Dry Areas
- Max 15 characters

PAP0000594#8.2 FIELD EQUIPMENT, ON PROCESS PIPING AND MACHINE FRAME

#8.2



- Acid-proof steel, laser printing, black text
- Acid-proof steel wire or cable tie (EN 1.4404 / AISI 316 L)

Usage:

- Hot or Wet Areas
- Max 15 characters

PAP0000594#9.1 CABLE MARKINGS



- Phoenix Contact Cable Markers **KMK2** (or **KMK4** for extralong text) are used instead of cable plates: **VAL0134728 KMK2** and **VAL0255656 KMK4**

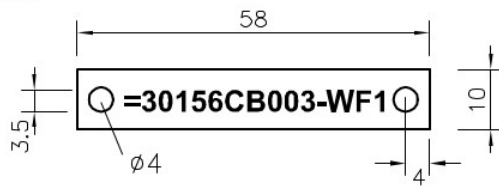
- White paper inside KMK2 or KMK4 pocket, black text
- Fastening with plastic cable tie

Usage:

- Dry Areas

PAP0000594#9.3 CABLE PLATES

#9.3



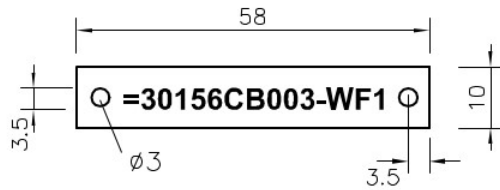
- Acid-proof steel, laser printing
- Steel plate, black text
- Acid-proof steel wire or cable tie (EN 1.4404 / AISI 316 L)

Usage:

- Hot or Wet Areas
- Max 18 characters

PAP0000594#10.1 PIPE PLATES

#10.1



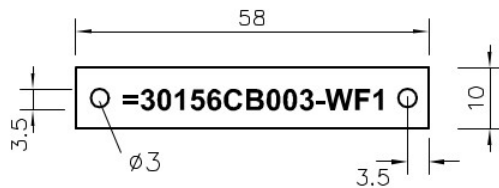
- Acid-proof steel, laser printing, black text
- Fastening with acid-proof steel wire or cable tie (EN 1.4404 / AISI 316 L) or
- Acid-proof screw or drive rivet fastening.

Usage:

- All pipes
- Max 18 characters

PAP0000594#10.2 PIPE PLATES

#10.2



- Paper printout protected with plastic. (Partex PM cable marking)
- White background, black text
- Fastening with cable tie

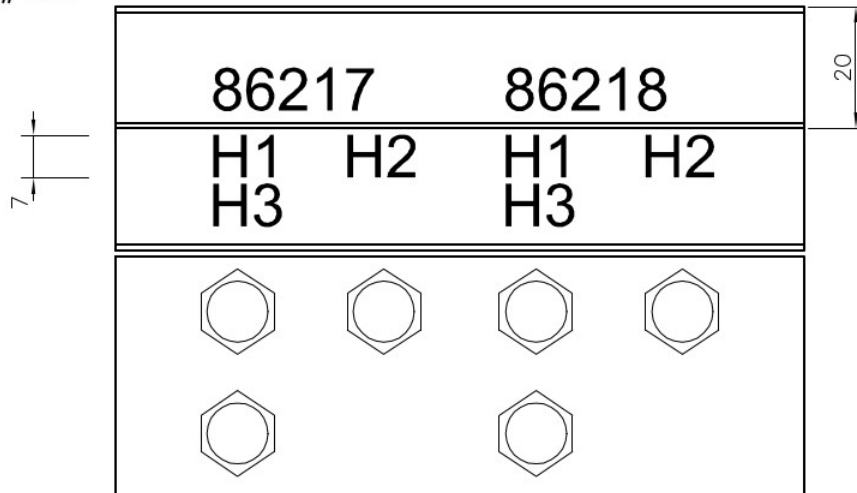
Usage:

- Inside valve panels
- Max 18 characters

PAP0000594#10.3 PIPE PLATES

Aluminum sheet with stamps, e.g. pipelines of circuits =30186GHZ217 and =30186GHZ217

#10.3



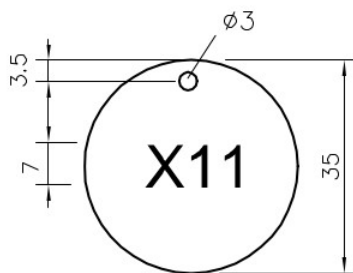
- Aluminum sheet. Height 20 mm.
- Stamped, height of text 7 mm.

Usage:

- Hydraulic Units, Pipes at Valve Panels

PAP0000594#10.4 PIPE PLATES

#10.4

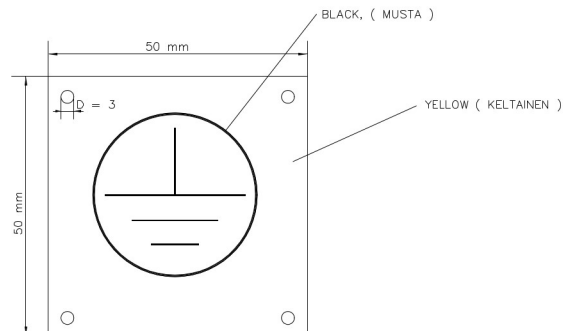


- Acid-proof steel, laser printing, black text
- Coupling code marked on the plate.
- Fastening with acid-proof steel wire (EN 1.4404 / AISI 316 L)

Usage:

- Hydraulic Units, Markings of Hoses Equipped with Quick Couplings

GROUNDING LABEL FOR MACHINE FRAMES AND HYDRAULIC UNITS:



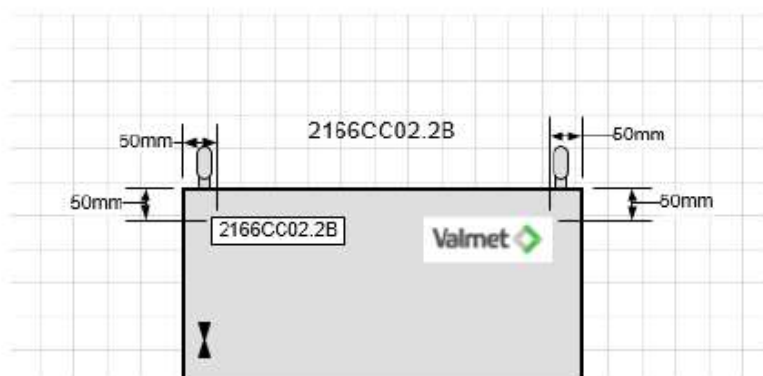
- Acid-proof steel, laser printing, black text, yellow background
- Acid-proof screw fastening

Usage:

- The grounding label is used to indicate the right point for field grounding cable on machine frames and hydraulic units

VALMET STICKER FOR ALL AUTOMATION CABINETS AND -BOXES:

Cabinet id marking plate is located on the left corner of the door (or in the middle of the door on the wider cabinets).



The default brand label is **GDI0004330 : VALMET Sticker 195x70 PVC Separate letters** in IP54 cabinets. Is located in the right corner of the door.

For small boxes/panels default brand label is **GDI0006293** 120x43 mm

PAP0000594 NAME PLATES:

See attached files.

VALMET GLOBAL NAME PLATES IN CADMATIC:

See attached files.

CONTROL EQUIPMENT IN DESKS AND CONTROL BOXES:

See attached files.